

UNIVERSITY INSTITUTE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BACHELOR OF ENGINEERING

CSE/IT/ THIRD SEMESTER

COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.3

Lecture No. 2.3.5

Topic : Virtual Memory and Memory Management Unit

Er. Shikha Atwal (E11186)

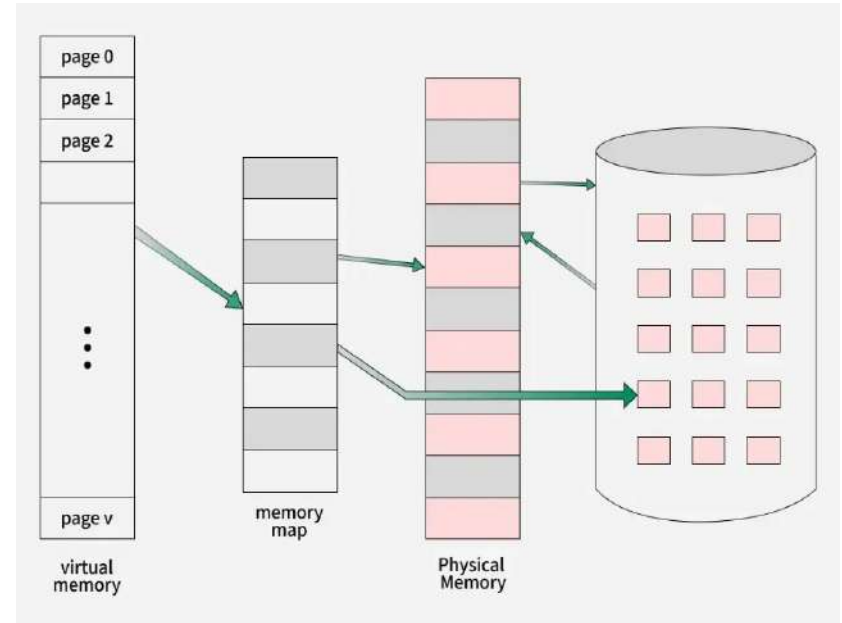
Assistant Professor

Learning Objectives

1. To understand the concept of virtual memory and how it allows execution of programs larger than physical memory.
2. To learn the role of the Memory Management Unit (MMU) in translating virtual addresses to physical addresses.
3. To analyze how virtual memory and MMU enable process isolation, efficient memory allocation, and paging mechanisms in modern operating systems.

Content

1. Virtual Memory
2. Types of Virtual memory
3. Paging
4. Segmentation
5. Segmented Paging
6. Memory Management Unit
7. Techniques used in MMU



Virtual Memory

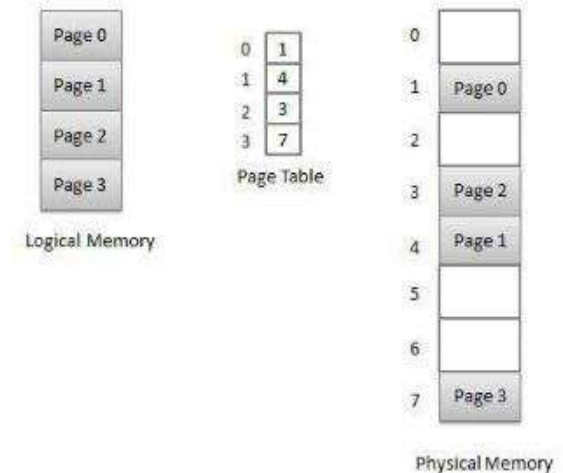
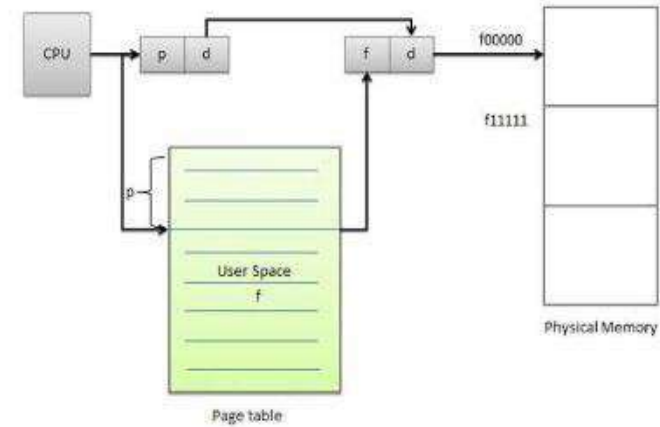
- Memory management technique that uses hard drive space as an extension of a computer's RAM.
- Allows to run programs that are larger than the physical memory.
- Creates an illusion for users and applications that there's a large, continuous memory space, even if the actual physical memory is limited.

Types of Virtual Memory (VM)

- VM is managed by the Memory Management Unit (MMU), which is often built into the CPU.
- CPU generates virtual addresses that the MMU translates into physical addresses.
- Two types of VM: **Paging and Segmentation**
- Paging and segmentation are processes by which data is stored to, then retrieved from a computer's storage disk.

Paging

- Memory management function that presents storage locations to the CPU as additional memory, called virtual memory.
- Physical memory is divided into blocks of the same size called pages.



paging table architecture.

Paging

- Page table stores the details of each page.
- Paging is when an active process requests data, the MMU retrieves corresponding pages into frames located in physical memory for faster processing.
- Address generated by CPU is divided into-
 - **Page number (p)** - index in a page table which contains base address of each page in physical memory.
 - **Page offset (d)** - page offset combined with base address to define the physical memory address.

Paging

- However, constant table lookups can slow down the MMU.
- A memory cache called the Translation Lookaside Buffer (TLB) stores recent translations of virtual to physical addresses for rapid retrieval.
- Many systems have multiple TLBs, which may reside at different locations, including between the CPU and RAM, or between multiple page table levels.

Issue in Paging

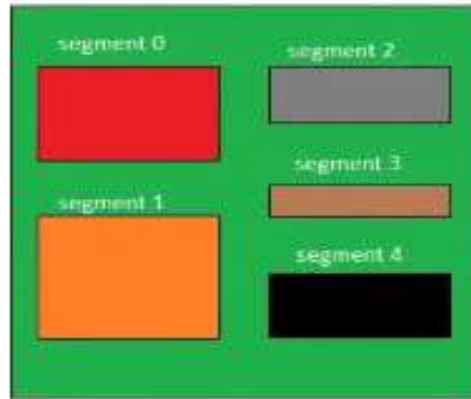
- There used to be a problem in older Windows deployments (95, 98, and ME).
- The problem was **internal (or hidden) fragmentation**.
- Unlike the serious external fragmentation of segmenting, internal fragmentation occurred if every frame is not the exact size of the page size.
- However, this is not an issue in modern Windows OS.

Segmentation

- It is a virtual process that creates variable-sized address spaces in computer storage for related data, called segments. This process speed retrieval.
- Each segment is a different virtual address space that directly corresponds to process objects.
- In this, CPU generates a logical address that contains the Segment number and segment offset.
- If the segment offset is less than the limit then the address is valid otherwise it throws miscalculation for invalid address.

Segmentation

Logical View of Segmentation

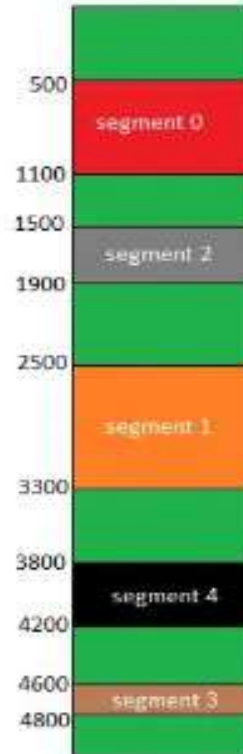


Logical Address Space

Segment Number

	base address	Limit
0	500	600
1	2500	800
2	1500	400
3	4600	200
4	3800	400

Segment Table



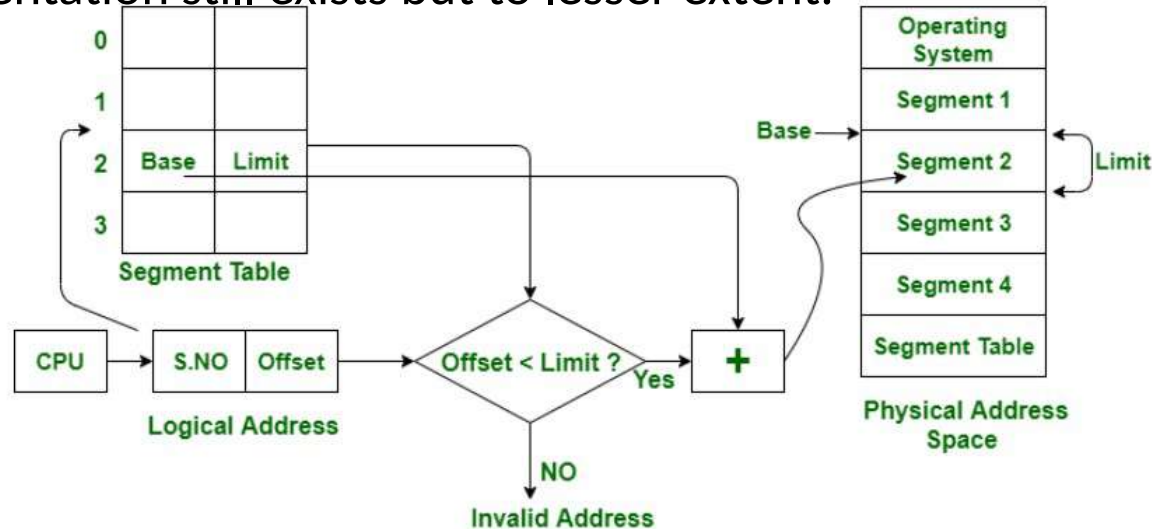
Physical Address Space

Segmentation

Segment number (s) - Index in a segment table which contains base address of each segment in physical memory and a limit of segment.

Segment offset (o) - size of segment and if less than limit then combined with base address to define the physical memory address.

External fragmentation still exists but to lesser extent.



Segmented Paging

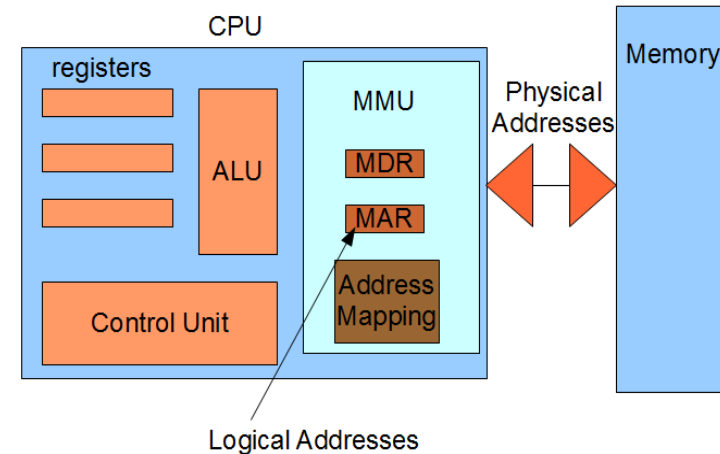
- Some modern computers use a technique called segmented paging.
- Main memory is divided into variably-sized segments, which are then divided into smaller fixed-size pages on disk.
- Each segment contains a page table, and there are multiple page tables per process.
- Each of the tables contains information on every segment page, while the segment table has information about every segment.

Memory Management Unit (MMU)¹³

- MMU is a key component of memory management hardware in computer architecture.
- It performs the essential task of translating virtual addresses generated by the CPU into physical addresses, allowing the system to access the correct memory location.
- The MMU works in conjunction with OS's memory management software to allocate and manage memory resources effectively.

Techniques used in MMU

- **Paging:** Dividing memory into fixed-size blocks called pages and mapping them in a virtual address space. It allows for efficient memory allocation and utilization by allocating pages based on demand.
 - **Segmentation:** Dividing memory into logical segments based on program structure and memory requirements. It provides a flexible memory allocation scheme but can lead to external fragmentation.
 - **Virtual Memory:** Technique that allows the execution of programs larger than the available physical memory. It uses disk-based storage to supplement the physical memory, transferring data between memory and disk as needed.
 - **Memory Protection:** The memory management hardware implements various memory protection mechanisms, such as access permissions and privilege levels, to ensure the security and integrity of memory.
-
- The diagram illustrates the interaction between the CPU and Memory. The CPU is shown as a large blue box containing several components: registers (three orange rectangles), an ALU (orange rectangle), a Control Unit (orange rectangle), and an MMU (light blue box). The MMU contains the MDR (orange rectangle), MAR (orange rectangle), and Address Mapping (brown rectangle). A double-headed arrow labeled 'Physical Addresses' connects the CPU to the Memory (blue box). A line points from the 'Address Mapping' component in the MMU to the text 'Address Mapping' in the original image.



Applications of the Topic

- Efficient Program Execution - Virtual memory enables running large programs by swapping parts in and out of physical memory, allowing multitasking.
- Process Isolation and Security - The MMU ensures that each process accesses only its allocated memory space, protecting system stability and preventing unauthorized access.
- Memory Paging and Swapping - OS use virtual memory with paging and swapping to manage memory efficiently without requiring continuous physical blocks.
- Virtualization and Cloud Computing - MMUs are crucial in supporting virtual machines by managing address translation and providing each VM with isolated virtual memory.

Summary of the Lecture

1. **Virtual memory** is a memory management technique that gives programs the illusion of having large, contiguous memory, even if the physical memory is limited.
2. It allows systems to run large applications and multiple processes simultaneously by storing parts of programs in disk storage and loading them into RAM as needed.
3. The **MMU** is a hardware component that handles the translation of virtual addresses to physical addresses, enabling features like **paging**, **segmentation**, and **process isolation**.
4. Together, virtual memory and MMU enhance **system performance**, **security**, and **flexibility** in modern computing environments.

Next Lecture

- **Input-Output Organization**
- **Interrupts**
- **Modes of Transfer**
- **I/O Processor**

Quiz

Q1. Which of the following best describes virtual memory?

- A. A type of physical memory inside the CPU
- B. Memory that can only be used by the operating system
- C. An illusion of a large memory space created using disk storage**
- D. A portion of RAM dedicated to the BIOS

Q2. A system uses 32-bit virtual addresses and has a page size of 4 KB. How many pages are there in the virtual address space?

- A. 2^{10}
- B. 2^{12}
- C. 2^{20}**
- D. 2^{32}

Q3. A computer has 1 GB of physical memory and a page size of 8 KB. How many page frames are available in physical memory?

- E. 2^{20}
- F. 2^{17}**
- G. 2^{19}
- H. 2^{14}

References/ Articles/ Videos

Text Books:

- Carl Hamacher, Zvonko Vranesic, Safwat Zaky, and Naraig Manjikian, 'Computer Organization and Embedded Systems', Sixth Edition, McGraw-Hill Education, 2012.
- M. Morris Mano, 'Computer System Architecture', Third Edition, Pearson Education, 2023.

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- John P. Hayes, 'Computer Architecture and Organization', Third Edition, McGraw-Hill Education, 1998.
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- Vincent P. Heuring, and Harry F. Jordan, 'Computer Systems Design and Architecture', Second Edition, Pearson Education, 2003.

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References Websites:

- <https://www.gues10.com/p/10067/what-is-virtual-memory-explain-the-role-of-pagin-1/>
- <https://www.techtarget.com/whatis/definition/memory-management-unit-MMU>
- [What Is Virtual Memory In Computer Architecture Explained](#)
- [VIRTUAL MEMORY MSC-IT](#)

Video Links:

- https://www.youtube.com/watch?v=6c-mOFZwP_8
- <https://youtu.be/30P73tWmU0s?si=NnPX3qjL64gU1veh>
- <https://youtu.be/dz9Tk6KCMIQ?si=sqQH6iaW1NFCFDu>
- <https://youtu.be/I2TbCGNv1xQ?si=YITcOfDJb0Ndmy6j>



Faculty-curated videos, NPTEL,²¹ Coursera, LinkedIn, or other relevant learning resources

https://youtu.be/TF_KP4SMIfM

https://en.wikipedia.org/wiki/Memory_management_unit

https://onlinecourses.nptel.ac.in/noc23_cs67/preview

<https://www.youtube.com/watch?v=FcLFzIJfPwQ>

https://www.youtube.com/watch?v=ujoJ7J_l9cY

<https://www.youtube.com/watch?v=AKGtJAi4wGo>

<https://www.coursera.org/learn/codio-intro-to-operating-systems-2-memory-management>

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Unit No. 2

Chapter No. 2.3

Lecture No. 2.3.4

Topic : Cache memory, Mapping Functions, Replacement Algorithms, Write Policies

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

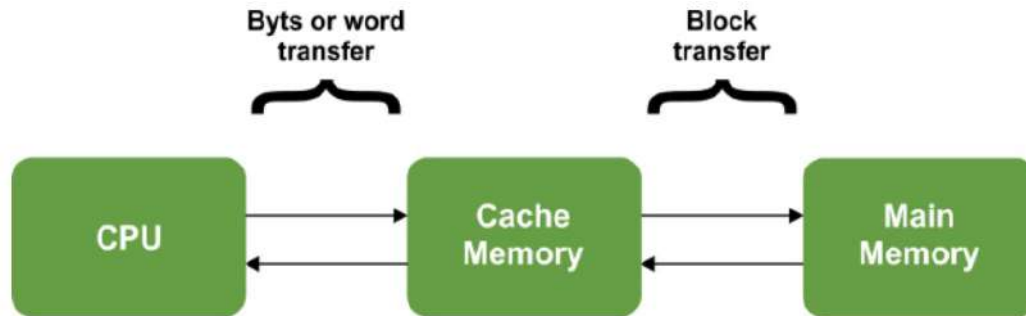
1. To understand the purpose and operation of cache memory in improving system performance by reducing average memory access time.
2. To learn different cache mapping techniques (direct, associative, set-associative) and replacement algorithms (FIFO, LRU, Optimal).
3. To analyze various write policies (write-through, write-back) and their impact on memory consistency, performance, and design trade-offs.

Content

1. Cache memory
2. Levels of Cache memory
3. Locality of Reference
4. Mapping Functions
5. Replacement Algorithms
6. Write Policies

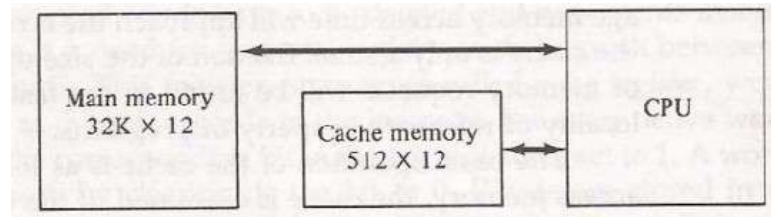
Cache Memory

- Stores the contents of the main memory that are used frequently by CPU
- Easy and faster access by the processor as whenever it needs to access memory, it first checks the cache memory.
- If not found in cache, then the CPU moves into the main memory.



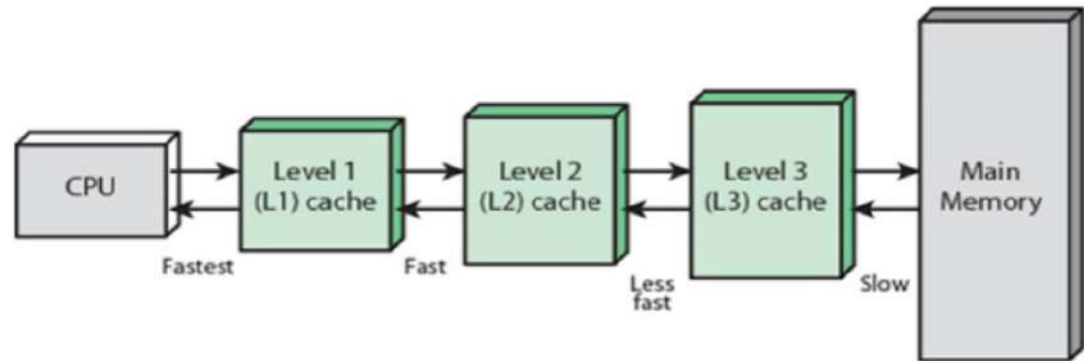
Principles of Cache Memory

- The main memory can store 32k words of 12 bits each.
- The cache is capable of storing 512 of these words at any given time.
- For every word stored, there is a duplicate copy in main memory.
- The CPU communicates with both memories. It first sends a 15-bit address to cache.
- If there is a hit, the CPU accepts the 12-bit data from cache. If there is a miss, the CPU reads the word from main memory and the word is then transferred to cache.



Levels of Cache Memory

- **L1 Cache (2KB to 64 KB):** Present inside the processor in a small amount inside every core of the processor separately.
- **L2 Cache (256 KB to 512 KB):** May present inside or outside the CPU. If not present inside the core, it can be shared between two cores depending upon the architecture.
- **L3 Cache (1 MB to 8MB):** Present outside the CPU and used to increase the performance of the L2 and L1 cache.



Locality of Reference

The two types of Locality of Reference are below:

1. **Temporal:** A recently executed instruction is likely to be executed again very soon. Ex- reuse of information in loops.
2. **Spatial:** Instructions in close proximity to a recently executed instruction are also likely to be executed soon. Ex- related data items (arrays) are usually stored together.

Cache Performance

On searching in the cache if data is found, a **cache hit** has occurred else if data is not found, a **cache miss** has occurred.

Hit Ratio = Number of hits / Total CPU references to memory
= Number of hits / (Number of hits + Number of misses)

Miss Ratio = Number of misses / Total CPU references to memory
= Number of misses / (Number of hits + Number of misses)

Miss Ratio = 1 - Hit Ratio

Cache Performance

Average Memory Access Time (t_{avg})

= hit ratio X cache access time + miss ratio X (cache access time + main memory access time)

= cache access time + miss ratio X main memory access time

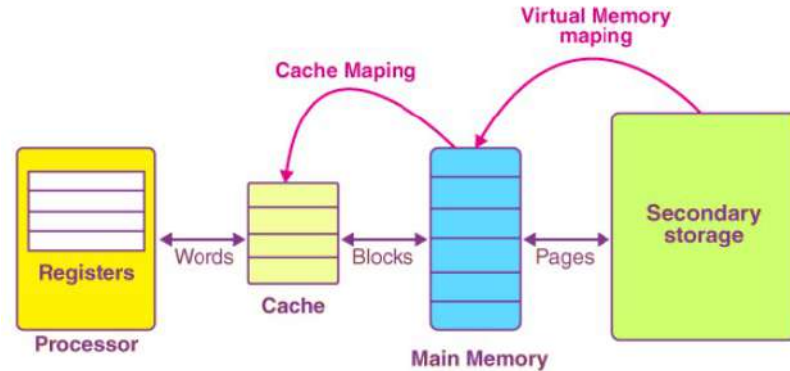
= Hit Time + Miss Ratio X Miss Penalty

Miss Penalty: Addition clock cycles to service the miss, the extra time needed to carry the favored information into cache from main memory in case of miss in cache.

Cache Mapping

Whenever a cache miss occurs,

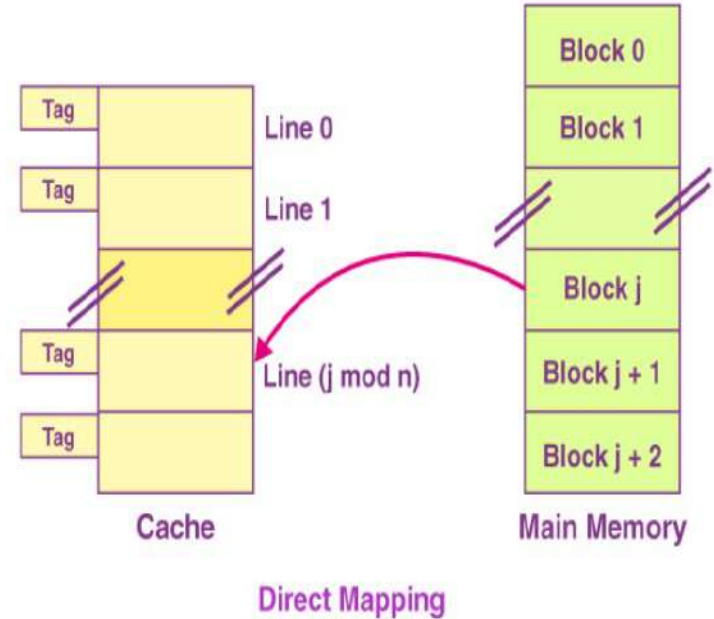
- The page consists of the required word that we need to map from the main memory.
- We can perform such a type of mapping to copy the data from main memory to cache using various different techniques of cache mapping.



Direct Mapping

- A certain block of the main memory would be able to map only to a certain line of the cache.
- Mapping can be done using following formula:

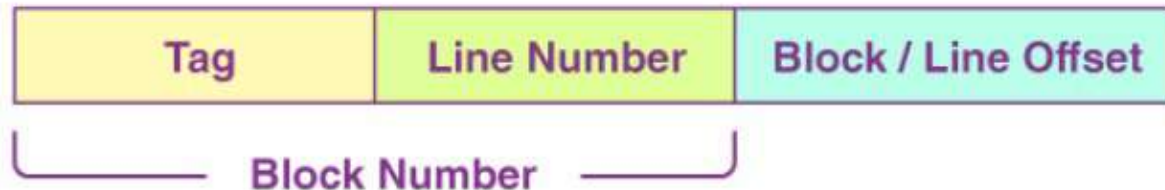
$$\text{Cache line number} = (\text{Address of the Main Memory Block}) \bmod (\text{Total number of lines in Cache})$$



Direct Mapping

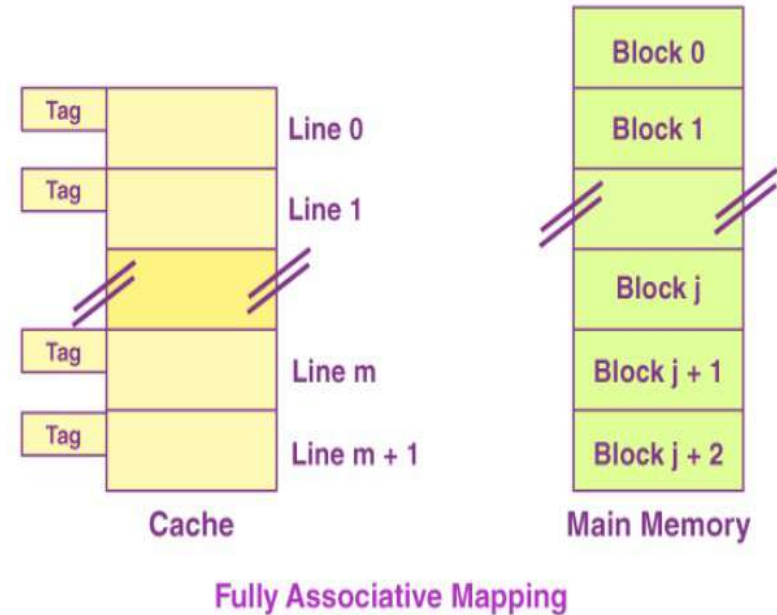
There is no requirement for a replacement algorithm.

The division of the physical address occurs as follows:



Fully Associative Mapping

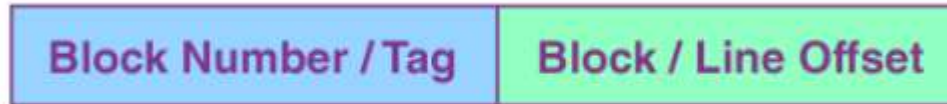
- The main memory block is capable of mapping to any given line of the cache that's available freely at that particular moment.
- In case all the cache lines are occupied, one of the blocks that exists already needs to be replaced.



Fully Associative Mapping

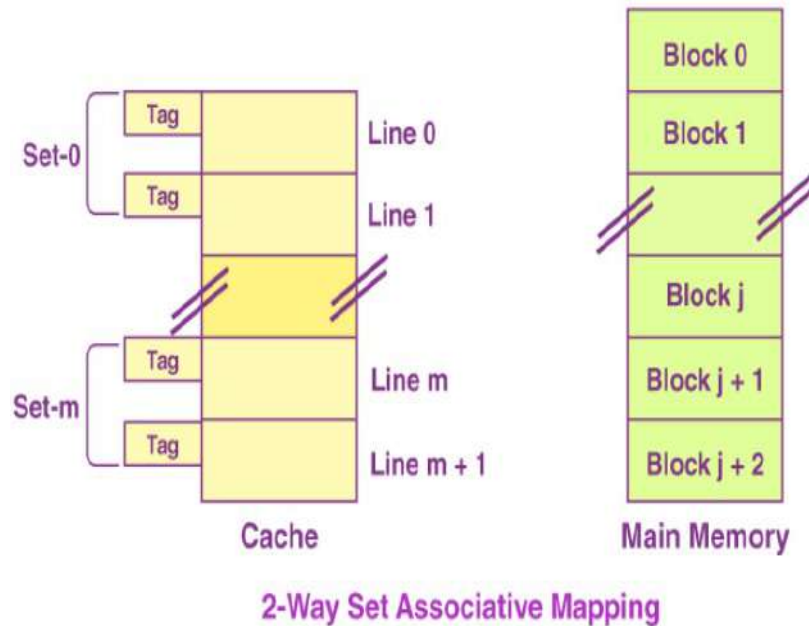
There will be a requirement of Replacement Algorithms.

The division of the physical address occurs as follows:



K-way Set Associative Mapping

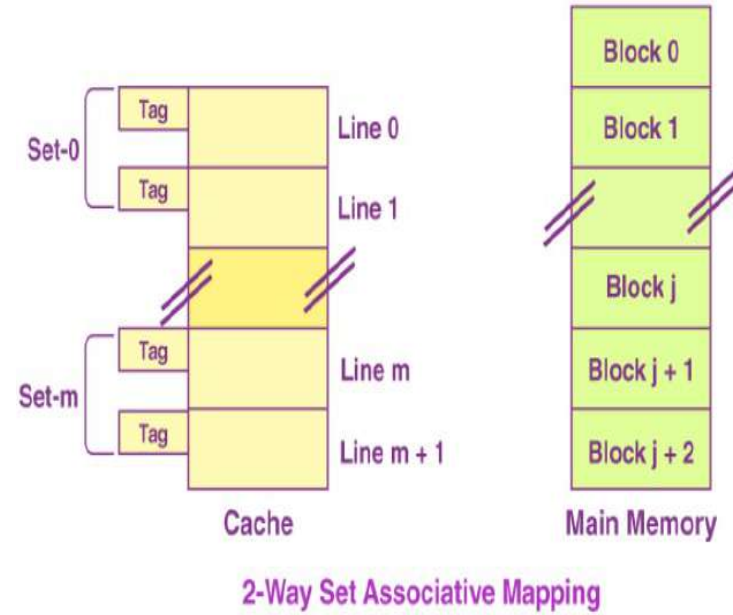
- The grouping of the cache lines occurs into various sets where all the sets consist of k number of lines.
- Any given main memory block can map only to a particular cache set.



K-way Set Associative Mapping

- However, within that very set, the block of memory can map any cache line that is freely available.
- The cache set to which a certain main memory block can map is given as follows:

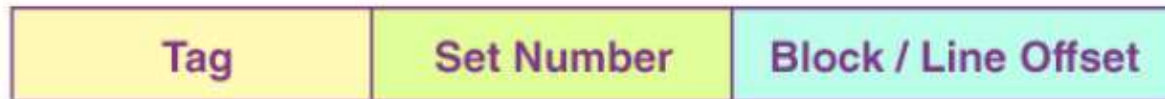
Cache set number = (Block Address) Modulo (Total Number of sets present in the Cache)



K-way Set Associative Mapping

In case all the available cache lines happen to be occupied, then one of the blocks that already exist needs to be replaced.

The division of the physical address occurs as follows:



Cache Replacement Algorithms

- Used when there is no available space in a cache to place a data.
- Needed to decide which page needs to be replaced when the new page comes in.
- Whenever a new page is referred to and is not present in memory, the **page fault** occurs and the OS replaces one of the existing pages with a newly one.

First In First Out (FIFO)

- The block which has entered first in the main memory will be replaced first.
- This can lead to a problem known as **Belady's Anomaly** which states that if we increase the no. of lines in cache memory the cache miss will increase which will increase the page fault.

First In First Out (FIFO)

Example: Let we have a sequence 7, 0 ,1, 2, 0, 3, 0, 4, 2, 3, and cache memory has 4 lines.

7	0	1	2	0	3	0	4	2	3
7	7	7	7		3		3		
	0	0	0		0		4		
		1	1		1		1		
			2		2		2		
m	m	m	m	H	m	H	m	H	H

6 page faults

Least Recently Used (LRU)

- The page which was not used for the largest period of time in the past will get replaced first.
- LRU is better than FIFO replacement as it can never exhibit Belady's Anomaly.
- It may require a sustainable hardware resource.

Least Recently Used (LRU)

Example: Let we have a sequence 7, 0 ,1, 2, 0, 3, 0, 4, 2, 3 and cache memory has 3 lines.

7	0	1	2	0	3	0	4	2	3
7	7	7	2		2		4		4
	0	0	0		0		0		3
		1	1		3		2		2
M	M	M	M	H	M	H	M	H	M

6 page faults

Optimal Replacement Algorithm

- The page that will not be needed for the longest period of time in the future will be replaced.
- It predicts which page will be accessed furthest ahead.
- Thus, the OS needs information on future memory references to implement it.
- However, it is not practical in real-world scenarios.

Optimal Replacement Algorithm

Example: Let we have a sequence 7, 0, 1, 2, 0, 3, 0, 4, 2, 3 and cache memory has 3 lines.

7	0	1	2	0	3	0	4	2	3
		1	1	1	3	3	3	3	3
	0	0	0	0	0	0	4	4	4
7	7	7	2	2	2	2	2	2	2
M	M	M	M	H	M	H	M	H	H

6 page faults

Least Frequently Used (LFU)

- Uses a counter to keep track of how often an entry is accessed.
- The entry with the lowest count is removed first.
- This method isn't used that often, as it does not account for an item that had an initially high access rate and then was not accessed for a long time.
- Similarly, we have **Most Frequently Used (MFU)** as well which is also not used often.

Random Replacement (RR)

- This randomly selects an object when it reaches maximum capacity.
- No reference or history of objects needed
- Very simple to implement.
- Used in ARM processors and the famous Intel i860.

Cache Write Policies

To save changes to main memory, there are two techniques involved:

1. **Write Through:** Every time an operation occurs, you update main memory as well as cache simultaneously. No risk of data loss.

2. **Write Back:** Saves data to cache only.

But at certain intervals or under a certain condition you would save data to the main memory. High probability of data loss.

Applications of the Topic

- Processor Performance Optimization to reduce latency and improve instruction execution speed.
- Embedded and Real-Time Systems help meet strict timing requirements and power constraints.
- Operating System Design and synchronization between main memory and cache in multi-tasking environments.
- Database and File System Caching to manage frequently accessed data efficiently.

Summary of the Lecture

1. Cache memory is a small, high-speed memory located close to the CPU that stores frequently accessed data to reduce the average memory access time.
2. To determine where data is placed in the cache, mapping functions such as direct mapping, associative mapping, and set-associative mapping are used.
3. When the cache is full, replacement algorithms like LRU (Least Recently Used), FIFO, and Optimal help decide which data to evict.
4. Additionally, write policies such as write-through and write-back define how and when data updates are propagated to main memory.
5. Together, these mechanisms significantly impact system performance, efficiency, and consistency.



Next Lecture

- **Virtual Memory**
- **Paging**
- **Segmentation**
- **Memory Management Unit**

Quiz

Q1. A cache has 64 blocks and uses direct mapping. The main memory has 1024 blocks. How many blocks of main memory can map to a single cache block?

- A. 8
- B. 16**
- C. 32
- D. 64

Q2. A system has a cache of 4 KB with a block size of 64 bytes. How many blocks are there in the cache?

- E. 8
- F. 16
- G. 32
- H. 64**

Q3. In a 2-way set associative cache with 128 sets and a block size of 32 bytes, what is the total size of the cache in bytes (excluding tag and metadata)?

- I. 8 KB
- J. 8 Kb
- K. 16 KB
- L. 16 Kb**

References/ Articles/ Videos

Text Books:

- Carl Hamacher, Zvonko Vranesic, Safwat Zaky, and Naraig Manjikian, 'Computer Organization and Embedded Systems', Sixth Edition, McGraw-Hill Education, 2012.
- M. Morris Mano, 'Computer System Architecture', Third Edition, Pearson Education, 2023.

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- [Differences between Associative and Cache Memory – GeeksforGeeks](#)
- <https://searchstorage.techtarget.com/definition/cache-algorithm>
- <https://www.includehelp.com/cso/types-of-cache-replacement-policies.aspx>

Video Links:

- https://youtu.be/wl6_dl4WjIY?si=Hoz7CndJ95pQ71Yz
- <https://youtu.be/OfqzoQ9Kw9k?si=K3S7xGMboveTzY7z>
- https://youtu.be/QZ_9Oe5E61Q?si=mslQJSaHwmd-Kbki
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- <https://youtu.be/5LmylpJcd9I?si=ljbnbbbzAkuldULz>

Faculty-curated videos, NPTEL,³³ Coursera, LinkedIn, or other relevant learning resources

- https://onlinecourses.nptel.ac.in/noc21_cs61/preview
- <https://www.coursera.org/learn/comparch>
- <https://www.youtube.com/watch?v=L9X7XXfHYdU&list=PLxCzCOWd7aiHMonh3G6QNKq53C6oNXGrX>
- <https://www.baeldung.com/cs/cache-write-policy>

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Unit No. 2

Chapter No. 2.3

Lecture No. 2.3.3

Topic : Auxiliary Memory, Associative Memory

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

1. To understand the role of auxiliary memory (e.g., hard disks, SSDs, optical storage) in storing large volumes of data permanently.
2. To learn the concept of associative (content-addressable) memory, which retrieves data based on content rather than address.
3. To analyze how auxiliary and associative memory support data backup, retrieval efficiency, and system performance in modern computing.

Content

1. Auxiliary Memory
2. Difference between Primary and Auxiliary memory
3. Uses of Secondary storage
4. Fixed and Removable Storage
5. Associative Memory

Auxiliary Memory

- Used for long-term storage of data and programs.
- Also known as external memory.
- Typically larger and slower than primary memory.
- Offer much larger storage capacity.



Difference between Primary and Auxiliary Memory

Primary Memory	Auxiliary Memory
Primary memory is directly accessed by the Central Processing Unit (CPU).	It is not accessed directly by the Central Processing Unit (CPU). Instead, data accessed from this memory is first loaded into Random Access Memory (RAM) and is then sent to the Processing Unit.
RAM provides a much faster-accessing speed to data than secondary memory. By loading software programs and required files into primary memory (RAM), computers can process data much more quickly.	This memory is slower in data accessing. Typically, primary memory is six times faster than auxiliary memory.
Primary memory, i.e. Random Access Memory (RAM) is volatile and gets completely erased when a computer is shut down.	This memory provides a feature of being non-volatile, which means it can hold on to its data with or without electrical power supply.

Uses of Auxiliary Memory

- **Portability:** Used to transfer the data from one device to another.
- **Permanent Storage:** RAM is volatile, so in order to secure the data permanently in the device, secondary storage devices are needed.



Fixed Storage

- Internal media device used to store data, and referred to as the Fixed disk drives or Hard Drives.
- Devices are literally not fixed, obviously, these can be removed from the system for repairing work, maintenance purposes, and also for an upgrade, etc.
- Types are Internal flash memory (rare), SSD (solid-state disk) units and Hard disk drives (HDD).

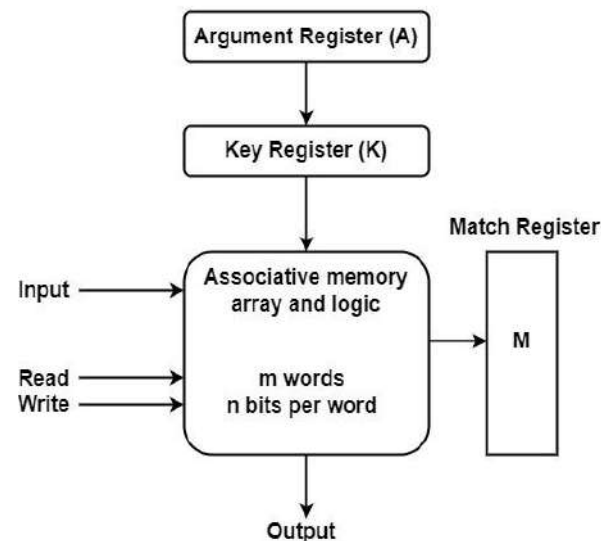
Removable Storage

- External media device used to store data, and referred to as the Removable Disks drives or the External Drives.
- Type of device that can be removed/ejected from a computer system while the system is running.
- Types are Optical discs, Memory cards, Floppy disks, Magnetic tapes, Disk packs, and Paper storage (punched tapes, punched cards).

Associative Memory

- Memory unit whose saved information can be recognized by the content of the information itself instead of by an address or memory location.
- Also known as **Content Addressable Memory (CAM)**.

Block Diagram of Associative Memory



Associative Memory

Step 1: Load the Argument Register (A)

A holds the data word (search pattern) that you want to compare with the contents of memory.

Step 2: Load the Key Register (K)

K determines which bits of A are used for the comparison.

$K[i] = 1$: Compare bit $A[i]$ with memory.

$K[i] = 0$: Ignore bit $A[i]$ (masking effect).

Step 3: Compare with All Memory Words in Parallel

The Associative Memory Array and Logic contains m words, each n bits long. The search logic performs bitwise parallel comparison between relevant bits of A and each stored word.

Associative Memory

Step 4: Store Comparison Result in Match Register (M)

Each bit in the M corresponds to a word in memory.

$M[i] = 1$: Word i matches the search pattern (based on K mask).

$M[i] = 0$: No match for word i .

Step 5: Output the Matched Word(s)

If one or more matches are found, the corresponding output word(s) are read from memory.

If needed, operations like read, write, or update can be performed on the matched word(s).

Applications of the Topic

- Virtual Memory Systems via TLB speeds up virtual-to-physical address translation.
- High-Performance Computing (HPC) through content-based lookups and caching.
- Associative memory is used in routing tables for fast packet forwarding; auxiliary memory stores routing logs and configurations.
- ROM (auxiliary memory) holds firmware, and associative memory assists in quick instruction decoding and address matching.

Summary of the Lecture

1. Auxiliary memory refers to external, non-volatile storage systems like hard drives and SSDs, designed to hold large volumes of data permanently.
2. It complements main memory by storing programs, files, and backups not actively in use.
3. Associative memory (or CAM) retrieves data by content rather than address.
4. It enables parallel searches, making it ideal for high-speed applications such as cache tag matching, virtual memory address translation, and network routing.

Next Lecture

- **Cache Memory**
- **Mapping Functions**
- **Replacement Algorithms**
- **Write Policies**

Quiz

Q1. Which of the following is an example of auxiliary memory?

- A. CPU Register
- B. Cache Memory
- C. Hard Disk**
- D. RAM

Q2. Associative memory is also known as:

- A. Cache memory
- B. Virtual memory
- C. Content Addressable Memory (CAM)**
- D. Secondary memory

Q3. Which of the following best describes associative memory operation?

- A. Data is accessed based on content**
- B. Data is accessed sequentially
- C. Data is accessed by memory address only
- D. Data is stored permanently

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COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.3

Lecture No. 2.3.2

Topic : Main Memory-RAM and ROM chips

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

1. To understand the structure and function of main memory, including the roles of RAM (volatile) and ROM (non-volatile).
2. To differentiate between various types of RAM (SRAM, DRAM) and ROM (PROM, EPROM, EEPROM) based on usage and characteristics.
3. To analyze how RAM and ROM chips are integrated into a system for data storage, program execution, and firmware access.

Content

1. Main Memory
2. RAM Chips
3. ROM Chips
4. Memory Address Map
5. Memory Connection to CPU

Main Memory

- Central storage unit in a computer system.
- Large and fast memory which is used to store programs and data during the run time operations.
- Technology used for this is based on semiconductor integrated circuits (ICs). The ICs are classified into two major units:
 1. RAM IC chips
 2. ROM IC chips

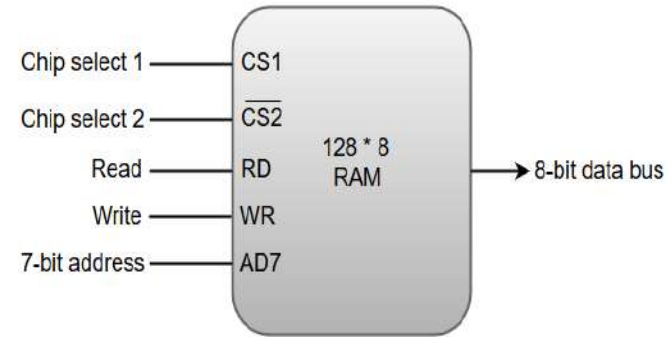
RAM Integrated Circuit Chips

- The RAM chips are classified into two possible operating modes: static and dynamic.
- Static RAM are made of flip-flops that store the binary information.
- Stored information is volatile, i.e. it remains valid as long as power is applied to the system.
- SRAM is easy to use and takes less time performing read and write operations.
- DRAM exhibits the binary information in the form of electric charges that are applied to capacitors.

RAM Integrated Circuit Chips

- A 128×8 RAM chip has capacity of 128 words of eight bits per word which requires a 7-bit address and an 8-bit bidirectional data bus.
- The data bus allows the transfer of data either from memory to CPU or from CPU to memory.
- The two chip select (CS) control inputs are for enabling the chip only when the microprocessor selects it.

Typical RAM chip:



RAM Integrated Circuit Chips

- The data bus is constructed using three-state buffers.
- The output generated by buffers can be placed in one of the three possible states which include a signal equivalent to logic 1, 0, or a high-impedance state.

CS1	$\overline{\text{CS2}}$	RD	WR	Memory function	State of data bus
0	0	x	x	Inhibit	High-impedance
0	1	x	x	Inhibit	High-impedance
1	0	0	0	Inhibit	High-impedance
1	0	0	1	Write	Input data to RAM
1	0	1	x	Read	Output data to RAM
1	1	x	x	Inhibit	High-impedance

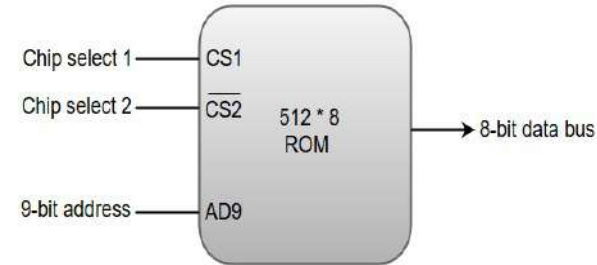
ROM Integrated Circuit Chips

- A portion of memory may be constructed with ROM chips.
- Used for keeping programs and data that are permanently resident in the computer.
- Needed for storing an initial program called a bootstrap loader.
- The primary function of the bootstrap loader program is to start the computer software operating when power is turned on.
- Similar organization as a RAM chip. However, a ROM can only perform read operation.

ROM Integrated Circuit Chips

- Data bus can only operate in an output mode.
- The nine address lines in the ROM chip specify any one of the 512 bytes stored in it.
- The two CS inputs must be CS1=1 and CS2=0 for the unit to operate. Otherwise, the data bus is said to be in a high-impedance state.

Typical ROM chip:



- The interconnection between CPU and memory is established from the knowledge of the size of memory required and the type of ROM and RAM chips available.
- The addressing of memory can be established by means of a table, that specify the memory address assigned to each chip, called the Memory address map.

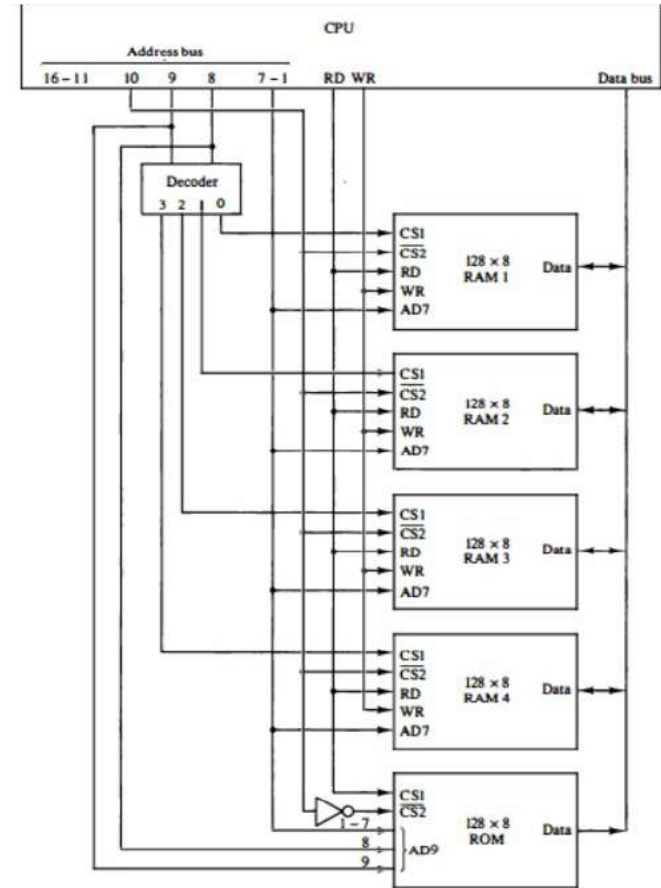
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Memory Connection to CPU

- The configuration is for memory capacity of 512 bytes of RAM and 512 bytes of ROM.
- Each RAM receives the 7 low-order bits of the address bus to select one of 128 possible bytes.
- The particular RAM chip selected is determined from lines 8 and 9 in the address bus. This is done through a 2 X 4 decoder whose outputs go to the CS1 inputs in each RAM chip.
- Thus, when address lines 8 and 9 are equal to 00, the first RAM chip is selected. When 01, the second RAM chip is select, and so on.

Memory Connection to CPU

- Address bus lines 1 to 9 are applied to the input address of ROM without going through the decoder.
- The data bus of the ROM has only an output capability, whereas the data bus connected to the RAMs is bidirectional.



Applications of the Topic

- System Memory for Program Execution (RAM) enables fast CPU access and efficient multitasking.
- Firmware Storage (ROM) initialize hardware during system startup.
- Embedded Systems like washing machines, routers, and smart appliances.
- Graphics and Game Performance (High-Speed RAM) used in GPUs for rendering graphics and improving gaming or multimedia performance.

Summary of the Lecture

1. Main memory is a vital component in computer systems, consisting of RAM and ROM.
2. RAM is volatile, high-speed memory used for temporarily storing data and instructions currently being processed by the CPU. It enables fast read/write operations essential for program execution and multitasking.
3. ROM is non-volatile and permanently stores critical programs like firmware and the BIOS, which are necessary for booting and basic hardware control.

Next Lecture

- **Auxiliary memory**
- **Associative memory**
- **Cache memory**
- **Virtual memory**

Quiz

Q1. Which of the following types of memory is volatile and used for temporary data storage during program execution?

- A. ROM
- B. DRAM**
- C. EEPROM
- D. PROM

Q2. What is the primary function of ROM in a computer system?

- A. Store temporary application data
- B. Execute multitasking operations
- C. Store firmware and boot instructions**
- D. Serve as cache memory

Q3. Which of the following types of ROM can be erased and reprogrammed using electrical signals?

- A. PROM
- B. EPROM
- C. SRAM
- D. EEPROM**

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COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.3

Lecture No. 2.3.1

Topic : Memory Organization-Memory Hierarchy

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

1. To explain the purpose of the memory hierarchy and the principle of locality.
2. To differentiate between the characteristics of RAM, ROM, cache, and storage.
3. To analyze the trade-offs between memory cost, speed, and capacity.

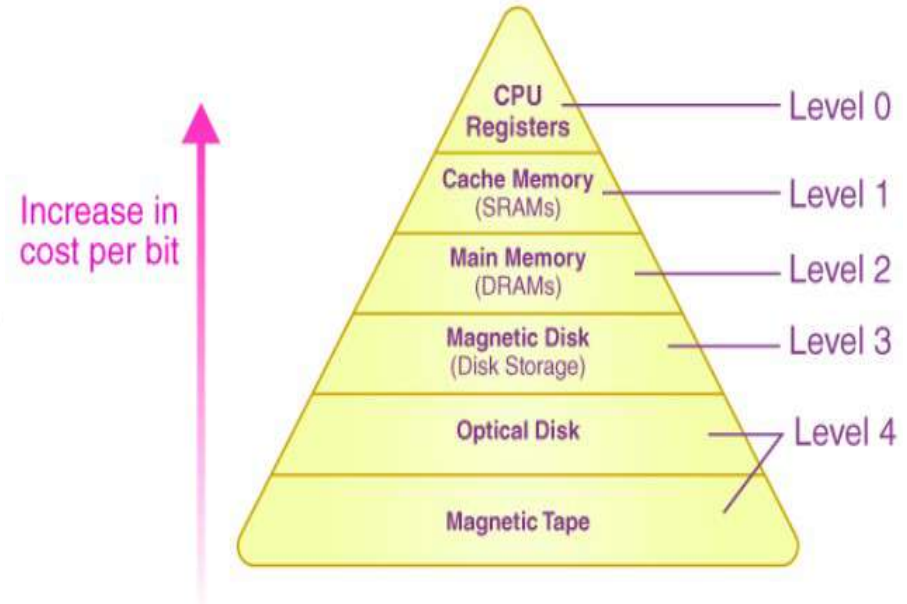
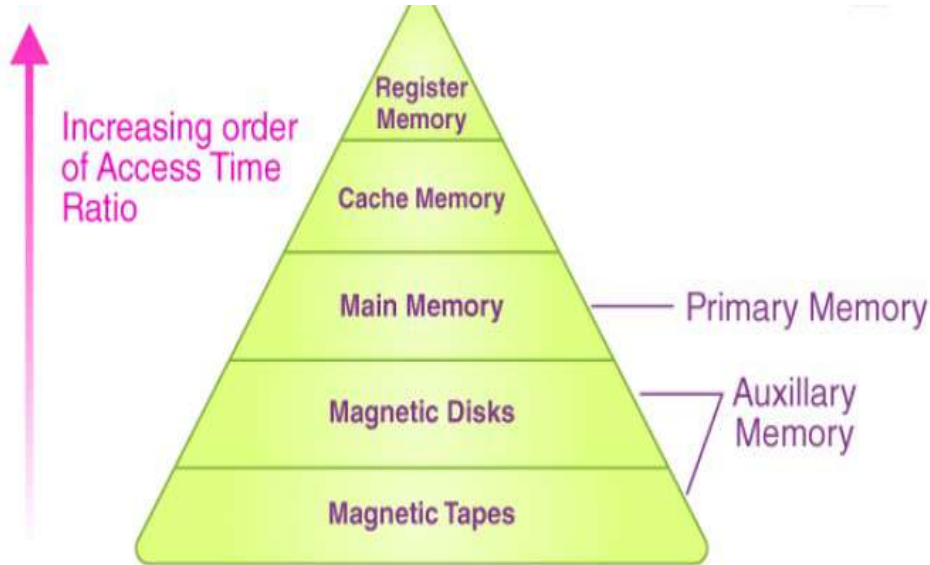
Content

1. Memory Organization
2. Memory Hierarchy
3. Characteristics of Memory hierarchy
4. Characteristics of Memory
5. Types of memory

Memory Organization

- Memory is the group of circuits used to store data.
- Size of the memory chip is denoted as the number of locations times the number of bits in each location.
- Data transfer between the memory and the processor takes place through two registers called MAR (Memory Address Register) and MDR (Memory data register).

Memory Hierarchy

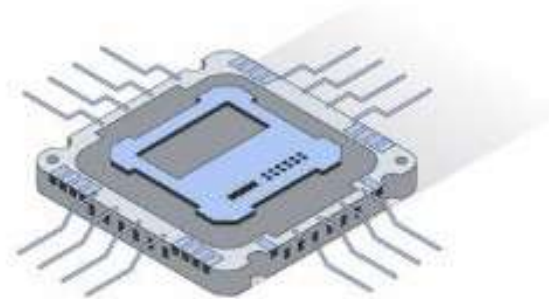


Memory Hierarchy

In computers, the memory hierarchy primarily includes the following:

1. Registers

- Usually an SRAM or static RAM in the computer processor
- Used to hold the data word typically 64 bits or 128 bits.
- A majority of the processors make use of a status word register and an accumulator.



Memory Hierarchy

2. Cache Memory

- Holds information that is used frequently from the main memory.
- In case the processor has a single-core, it will rarely have multiple cache levels.



Memory Hierarchy

3. Main Memory

- CPU's memory unit that communicates directly.
- Primary storage unit of a computer system.
- Very fast and a very large memory that is used for storing the information throughout the computer's operations.
- Made up of ROM as well as RAM.



Memory Hierarchy

4. Magnetic Disks

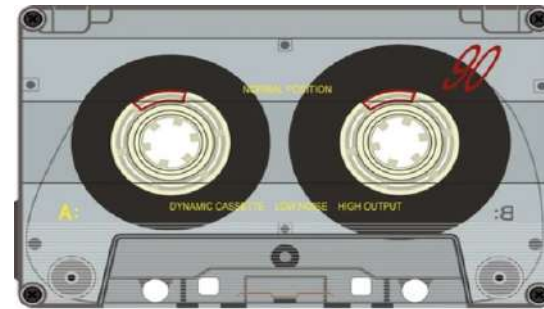
- Circular plates that's fabricated with plastic or metal with a magnetized material.
- Two faces of a disk are frequently used
- Many disks can be stacked on a single spindle by read/write heads that are obtainable on every plane.



Memory Hierarchy

5. Magnetic Tape

- Normal magnetic recording designed with a slender magnetizable overlay that covers an extended, thin strip of plastic film.
- Used mainly to back up huge chunks of data.
- The actual access time of a computer memory would be slower.



Characteristics of Memory Hierarchy¹⁰

1. **Capacity:** Total volume of data that a system's memory can store.
2. **Access Time:** Time interval present between the request for read/write and the data availability.
3. **Performance:** The large difference in the system's access time results in the system's lower performance, and thus, enhancement was required. Because of this enhancement, the system's performance increased.
4. **Cost per bit:** It increases as one moves from the bottom to the top in the Memory Hierarchy, i.e. External Memory is cheaper than Internal Memory.

Characteristics of Memory

- The storage capacity varies for different types of memories in computers.
- Unit of transfer measures the transfer rate of bits that can be either read or written in and out of memory devices.
- The internal memory in a computer is built within it, whereas external memory is present as a separate storage device such as USB or disk.
- There are four methods to access memory, including DMA, Associative Access Method, Sequential Access Method, and Random Access Method.
- Access time is the total time taken by memory devices to perform read and write operations.

Characteristics of Memory

Level	Register	Cache memory	Primary memory	Secondary memory
Bandwidth	4k to 32k MB/sec	800 to 5k MB/sec	400 to 2k MB/sec	4 to 32 MB/sec
Size	Less than 1KB	Less than 4MB	Less than 2 GB	Greater than 2 GB
Access time	2 to 5 nsec	3 to 10 nsec	80 to 400 nsec	5 ms
Managed by	Compiler	Hardware	Operating system	OS or user

Types of Memory

1. Primary Memory:

- Communicates directly with the CPU, cache and auxiliary memory.
- Allows immediate access from temporary memory slots or other storage locations.
- Two types: RAM and ROM.

1.1 RAM

- Hardware that temporarily stores data and programs.
- Faster part of the main memory which can be directly accessed by the CPU.
- Reads and writes programs until the computer is switched on.
- Two types: DRAM and SRAM.

Types of Memory

- **DRAM (Dynamic Random-Access Memory):** It is a type of RAM that is used for dynamic data storage. Every cell in DRAM consists of one-bit information.



- **SRAM (Static Random-Access Memory):** This type of RAM stores static data in memory which remains active until there is a power supply. Same sized SRAM chip holds less data than DRAM.



Types of Memory

1.2 ROM

- Read-Only Memory (ROM) is a permanent and non-volatile storage type.
- Only reads the stored information, but it does not have the capability to modify or write.



Types of Memory

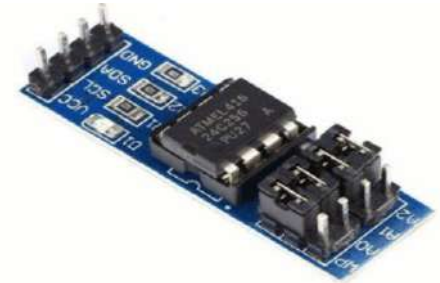
ROM is of the following five types:

- **MROM (Mask Read Only Memory):** Oldest ROM whose data is pre-configured via integrated circuit manufacture during the time of manufacturing.
- **PROM (Programmable Read Only Memory):** Digital ROM which only once allows writing any information or program using a special PROM programmer or burner device.
- **Flash ROM:** Programmed or written in smaller units called sectors or block.



Types of Memory

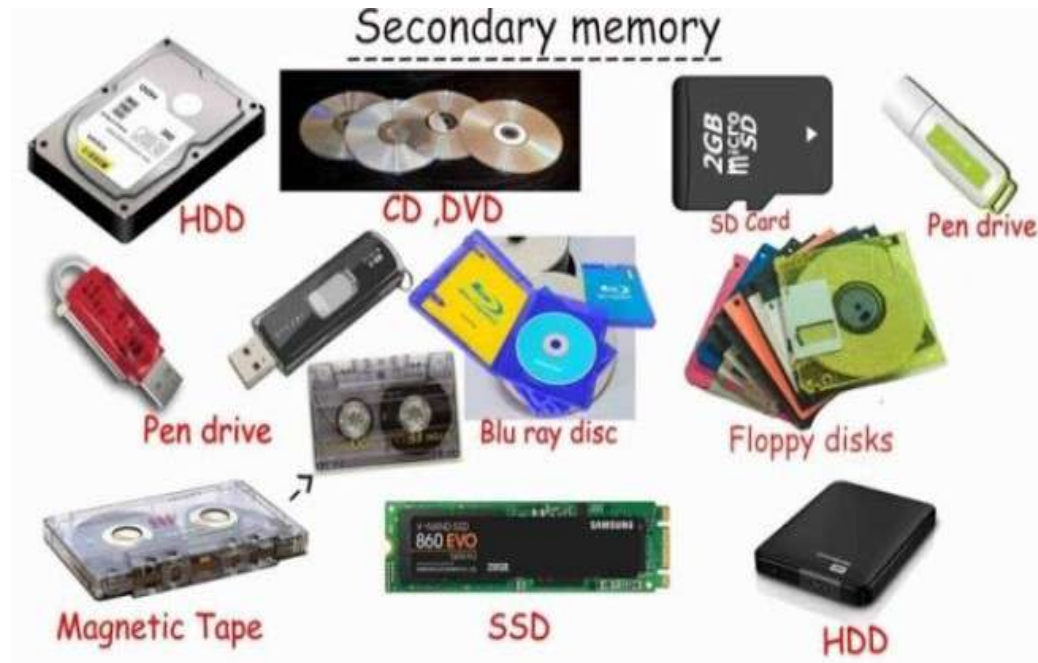
- **EPROM (Erasable Programmable Read Only Memory):** Data can be erased as well as reprogrammed only once. It can store data for a minimum of 10-20 years. To erase and reprogram EPROM, the user needs to pass UV light for 40 minutes. Post this, the data can be recreated.
- **EEPROM (Electrically Erasable Programmable ROM):** This allows data to be erased using a high-voltage electrical charge. After this, it can be reprogrammed up to thousands of times.



Types of Memory

2. Secondary Memory

- Permanent type of memory in computer that holds a large amount of data.
- Directly accessible by the CPU and is available as external devices.
- Cheaper but slower than primary memory.
- Example: Hard disk, Compact Disc (CD), Pen Drive



Types of Memory

3. Cache Memory

- Small-sized chip-based memory lies between the main memory and CPU.
- Very close to the chip of CPU.
- High-performance and temporary type of memory.
- Three types: L1, L2, and L3 cache.



Applications of the Topic

- Performance Optimization in CPUs by keeping the most frequently used data close to the processor.
- Cost-Effective Memory Design to balance cost and performance.
- Virtual Memory Management in Operating Systems by treating secondary storage as an extension of RAM to allow large program execution.
- Efficient Resource Allocation in Cloud and HPC Systems to optimize access time and data availability in large-scale distributed environments.

Summary of the Lecture

1. Memory hierarchy is a structured arrangement of storage types in a computer system, organized by speed, cost, and capacity.
2. This enables systems to access frequently used data quickly.
3. It stores less-used data in cheaper, larger storage.
4. Systems achieve a balance between performance and cost, ensuring that data is accessed as quickly and economically as possible.

Next Lecture

- **Main Memory-RAM and ROM chips**
- **Auxiliary Memory**
- **Associative Memory**

Quiz

Q1. Which of the following represents the correct order of memory hierarchy from fastest to slowest?

- A. Cache → Registers → RAM → Hard Disk
- B. Registers → Cache → RAM → Hard Disk**
- C. Hard Disk → RAM → Cache → Registers
- D. Registers → RAM → Cache → Hard Disk

Q2. Which level of the memory hierarchy typically has the highest storage capacity?

- A. Register
- B. Cache
- C. Main Memory (RAM)
- D. Hard Disk**

Q3. What is the primary benefit of having multiple levels in the memory hierarchy?

- A. Reduces instruction size
- B. Increases the number of registers
- C. Balances speed, cost, and storage capacity**
- D. Eliminates the need for RAM

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Unit No. 2

Chapter No. 2.2

Lecture No. 2.2.3

Topic : Comparative Study of Hardwired and Micro-programmed Control Unit

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

1. To understand the structural and operational differences between hardwired and microprogrammed control units in CPUs.
2. To evaluate the performance, flexibility, and complexity trade-offs involved in choosing one control unit type over the other.
3. To analyze real-world applications where each control unit type is preferred based on speed, cost, and upgradability.

Content

1. Comparative study of Hardwired and Micro-programmed Control Unit
2. Comparison based on factors

Differences between Hardwired and Micro-programmed Control Unit³

Hardwired Control Unit	Microprogrammed Control Unit
With the help of a hardware circuit, we can implement the hardwired control unit. In other words, we can say that it is a circuitry approach.	While with the help of programming, we can implement the micro-programmed control unit.
The hardwired control unit uses the logic circuit so that it can generate the control signals, which are required for the processor.	The micro-programmed CU uses microinstruction so that it can generate the control signals. Usually, control memory is used to store these microinstructions.
In this CU, the control signals are going to be generated in the form of hard wired. That's why it is very difficult to modify the hardwired control unit.	It is very easy to modify the micro-programmed control unit because the modifications are going to be performed only at the instruction level.
In the form of logic gates, everything has to be realized in the hardwired control unit. That's why this CU is more costly as compared to the micro-programmed control unit.	The micro-programmed control unit is less costly as compared to the hardwired CU because this control unit only requires the microinstruction to generate the control signals.

Differences between Hardwired and⁴ Micro-programmed Control Unit

Hardwired Control Unit	Microprogrammed Control Unit
The complex instructions cannot be handled by a hardwired control unit because when we design a circuit for this instruction, it will become complex.	The micro-programmed control unit is able to handle the complex instructions.
Because of the hardware implementation, the hardwired control unit is able to use a limited number of instructions.	The micro-programmed control unit is able to generate control signals for many instructions.
Difficult to modify as the control signals that need to be generated are hard wired	Easy to modify as the modification need to be done only at the instruction level
More costlier as everything has to be realized in terms of logic gates	Less costlier than hardwired control as only micro instructions are used for generating control signals
Only limited number of instructions are used due to the hardware implementation	Control signals for many instructions can be generated
Used in computer that makes use of Reduced Instruction Set Computers (RISC)	Used in computer that makes use of Complex Instruction Set Computers (CISC)



Comparison based on factors

Definition: Hardwired Control Unit is a unit that uses combinational logic units, featuring a finite number of gates that can generate specific results based on the instructions that were used to invoke those responses. Microprogrammed Control Unit is a unit that contains microinstructions in the control memory to produce control signals.

Speed: The speed of operations in Hardwired Control Unit is fast. The speed of operations in Microprogrammed Control Unit is slow because it requires frequent memory accesses.

Modification: To do modifications in a Hardwired Control Unit, the entire unit should be redesigned. In Microprogrammed Control Unit, modifications can be implemented by changing the microinstructions in the control memory. Therefore, Microprogrammed Control Unit is more flexible.

Handling Complex Instructions: Also, it is difficult for Hardwired Control Unit to handle complex instructions, but is easier for the Microprogrammed Control Unit to handle complex instructions.

Cost: Furthermore, Hardwired Control Unit are more costly to implement than a Microprogrammed Control Unit.



Comparison based on factors

Instruction Decoding: Moreover, it is difficult to perform instruction decoding in Hardwired Control Unit than in Microprogrammed Control Unit.

Instruction set Size: In addition to the above differences, the Hardwired Control Unit uses a small instruction set while the Microprogrammed Control Unit uses a large instruction set.

Control Memory: Also, there is no control memory usage in Hardwired Control Unit but, on the other hand, Microprogrammed Control Unit uses control memory.

Applications: Considering the applications, the Hardwired Control Unit is used in processors that use a simple instruction set known as the Reduced Instruction Set Computers (RISC). Microprogrammed Control Unit is used in processors based on a complex instruction set known as Complex Instruction Set Computer (CISC).

Applications of the Topic

- Processor Architecture Design based on application requirements.
- Compiler and Firmware Development in optimizing instruction execution based on control unit type.
- Instruction Set Design and Extension like hardwired for simple RISC-style ISAs, microprogrammed for complex CISC-style ISAs.
- Educational Tools and CPU Simulation to demonstrate trade-offs in control logic implementation through simulation and comparative analysis.

Summary of the Lecture

1. Hardwired and microprogrammed control units are two fundamental approaches to generating control signals in a CPU.
2. A hardwired control unit uses fixed combinational logic and is known for its speed and efficiency, making it ideal for simple, high-performance systems like RISC processors.
3. In contrast, a microprogrammed control unit uses a set of microinstructions stored in control memory, offering flexibility and ease of modification, which is beneficial for implementing complex instruction sets as seen in CISC architectures.
4. Understanding their differences enables better design choices based on system goals like speed, cost, upgradability, and instruction complexity.

Next Lecture

- **Memory Subsystem**
- **Types of memory**
- **Main Memory**
- **Auxiliary Memory**
- **Associative memory**

Quiz

Q1. Which of the following is an advantage of a microprogrammed control unit over a hardwired control unit?

- A. Faster execution
- B. Simpler hardware implementation
- C. Easier to modify and update**
- D. Requires no memory access

Q2. Hardwired control units are generally preferred in which type of architecture?

- A. CISC
- B. RISC**
- C. Virtual machines
- D. Microcontrollers with no ALU

Q3. Which of the following best explains a limitation of hardwired control units?

- A. Difficult to execute parallel instructions
- B. Cannot handle conditional instructions
- C. Cannot be updated without redesigning hardware**
- D. Slow execution of instructions

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CSE/IT/ THIRD SEMESTER

COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.2

Lecture No. 2.2.2

Topic : Micro-Programmed Control Unit

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

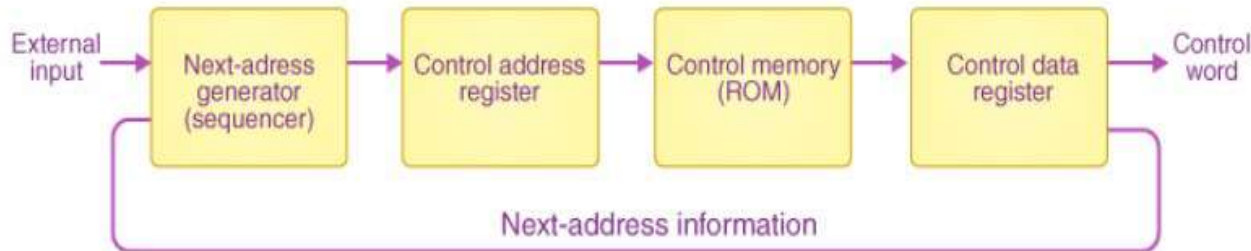
1. To understand the structure and working of a microprogrammed control unit, where control signals are generated using microinstructions stored in control memory.
2. To learn how control words, control memory, and microinstruction sequencing manage CPU operations.
3. To analyze the advantages of microprogrammed control in terms of flexibility, ease of modification, and support for complex instruction sets.

Content

1. Micro-Programmed Control Unit
2. Characteristics
3. Components
4. Step-wise Working
5. Timing Diagram
6. Typical Microinstruction format
7. Types of Micro-programmed CU

Micro-Programmed Control Unit

- The programming approach is used to implement it.
- A program made up of microinstructions is used to carry out a series of micro-operations.
- The control unit's control memory stores a microprogram composed of microinstructions.
- The creation of a set of control signals is dependent on the execution of a microinstruction.

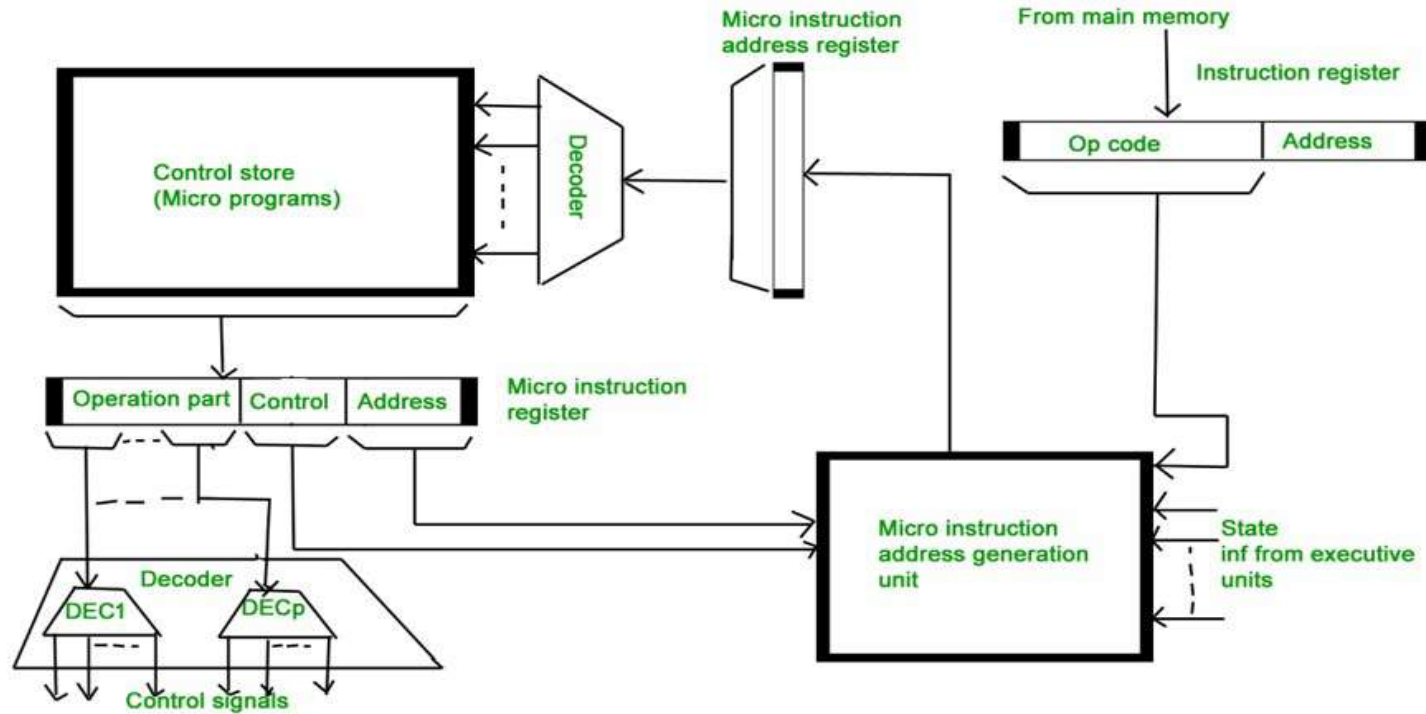


Components of Micro-Programmed

- **Control Memory (CM):** Stores microinstructions.
- **Control Address Register (CAR):** Holds the address of the next microinstruction.
- **Control Data Register (CDR):** Holds the current microinstruction fetched from CM.
- **Sequencer (Next Address Logic):** Decides the address of the next microinstruction.
- **Microinstruction Decoder:** Converts microinstruction into actual control signals.

Characteristics of Micro-Programmed CU

- The microinstruction address is specified in the control memory address register.
- All the control information is saved in CM, which is ROM.
- The microinstruction received from memory is stored in the control register.
- A control word in the microinstruction specifies one or multiple micro-operations for a data processor.
- The next address is calculated in the circuit of the next address generator and then transferred to CAR for reading the next microinstruction.
- Because it determines the sequence of addresses received from CM, the next address generator is also known as a microprogram sequencer.



Step-wise Working

Step 1: Initialize Control Unit

- CAR is loaded with the starting address of the microprogram for the current machine instruction (based on opcode).

Step 2: Fetch Microinstruction

- CAR sends address to Control Memory (CM).
- The microinstruction is fetched from CM and loaded into the Control Data Register (CDR).

Step 3: Decode and Execute Microinstruction

- The CDR outputs control signals to CPU components (ALU, registers, etc.).
- The specified micro-operations (like Load, Add, Shift) are executed.

Step-wise Working

Step 4: Determine Next Microinstruction

- Sequencer decides the next address:
 - Increment CAR (for sequential execution), or
 - Branch (based on flags or instruction requirements).

Step 5: Repeat Until End of Microprogram

- Steps 2-4 are repeated until the end of the microprogram (for the given instruction) is reached.
- The next machine instruction is then fetched from memory, and the cycle restarts.

Timing Diagram

- T0: $CAR \leftarrow$ Start Address (based on opcode)
- T1: $CDR \leftarrow CM[CAR]$ (Fetch microinstruction)
- T2: Decode CDR $\rightarrow$ Activate control signals (e.g., Load, Add)
- T3: Sequencer $\rightarrow$ Next CAR
- T4: Repeat fetch/decode until microprogram ends

Typical Microinstruction format

A typical microinstruction format may include:

[Control Signals | Next Address | Jump Control Bits]

Control Signals: Bits that activate specific CPU components (e.g., Load AC, Read Memory)

Next Address: Address of the next microinstruction

Jump Control: Bits to decide the next step (sequential, conditional, or jump)

Types of Micro-programmed CU

Based on the type of Control Word stored in CM, it is classified into two types:

1. Horizontal Micro-programmed control Unit:

- The control signals are represented in the decoded binary format that is 1 bit/CS.
- Example: If 53 Control signals are present in the processor then 53 bits are required. More than 1 control signal can be enabled at a time.
- It supports longer control word.

Types of Micro-programmed CU

2. Vertical Micro-programmed control Unit:

- The control signals are represented in the encoded binary format. For N control signals- $\log_2(N)$ bits are required.
- Supports shorter control words.
- Supports easy implementation of new control signals therefore it is more flexible.

Applications of the Topic

- Widely used in CISC processors (e.g., Intel x86) allowing easier support for rich instruction sets.
- Custom Instruction Set Extensions by updating microcode rather than redesigning hardware which is ideal for evolving computing needs.
- Fault-Tolerant and Maintainable Systems through firmware updates, useful in enterprise servers and mainframes.
- Educational Simulators and CPU Emulation to demonstrate instruction execution and emulate different CPU behaviors for teaching and experimentation.

Summary of the Lecture

1. A Microprogrammed Control Unit generates control signals through a sequence of microinstructions stored in control memory.
2. Each microinstruction activates a set of control signals and determines the address of the next instruction using a sequencer.
3. This architecture provides greater flexibility than hardwired control units, making it easier to implement, modify, and support complex instruction sets especially in CISC architectures.
4. Although slightly slower due to memory access, its adaptability makes it ideal for modern processors, firmware updates, and instructional simulations.

Next Lecture

- **Comparative study of Hardwired and Micro-programmed Control Unit**



Quiz

Q1. What is the main advantage of a microprogrammed control unit over a hardwired control unit?

- A. Faster instruction execution
- B. Lower hardware cost
- C. Easier to modify or update control logic**
- D. Uses fewer control signals

Q2. Microinstructions in a microprogrammed control unit are typically stored in:

- A. RAM
- B. Cache
- C. Register File
- D. Control Memory**

Q3. Which of the following components determines the next microinstruction address in a microprogrammed control unit?

- A. Program Counter
- B. Decoder
- C. Sequencer**
- D. Instruction Register

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COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.2

Lecture No. 2.2.1

Topic : Design of Control Unit - Hardwired CU

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

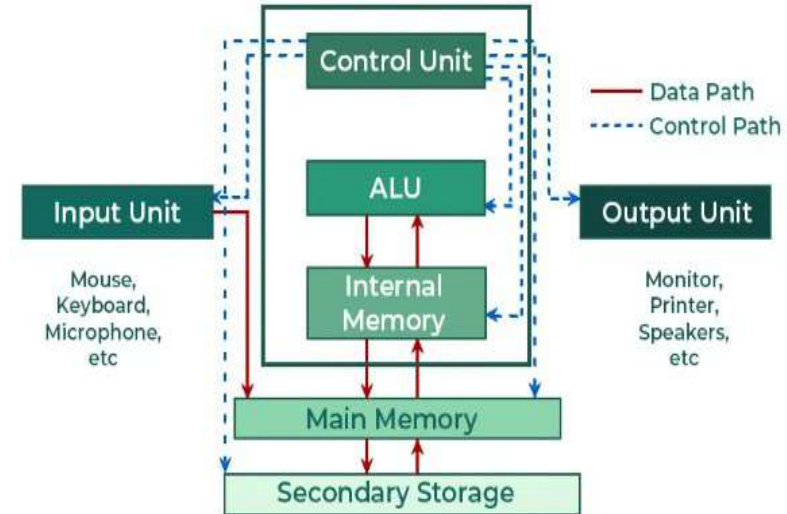
1. To understand the design and operation of a hardwired control unit, where control signals are generated through fixed logic circuits.
2. To learn how instruction execution is managed using combinational logic and sequential circuits.
3. To analyze the advantages and limitations of hardwired control units in terms of speed, complexity, and flexibility compared to microprogrammed control units.

Content

- Design of Control Unit
- Hardwired Control Unit

Design of Control Unit

- To execute an instruction, the control unit must generate the required control signal in the proper sequence.
- Two approaches used for generating the control signals are **Hardwired Control unit** and **Micro-programmed control unit**.

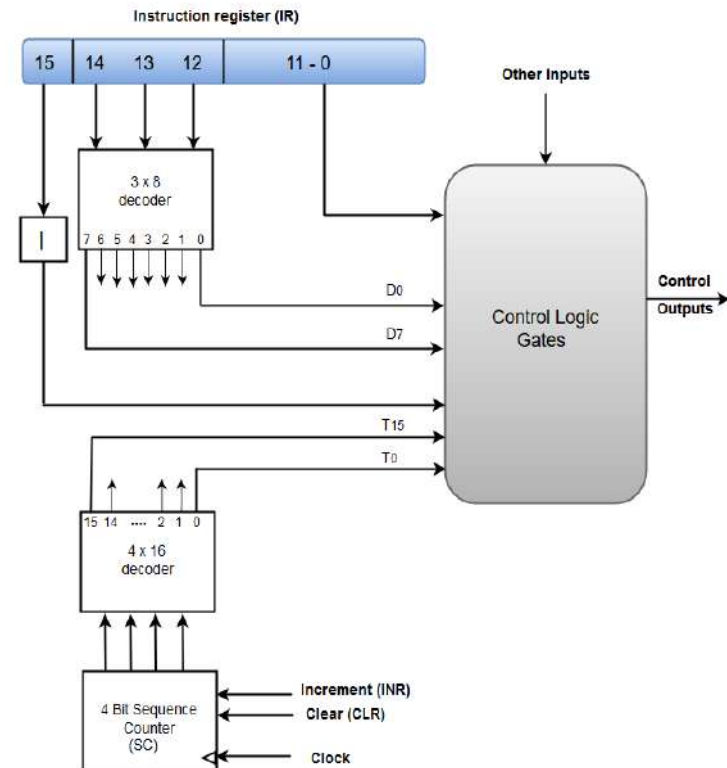


Hardwired Control Unit

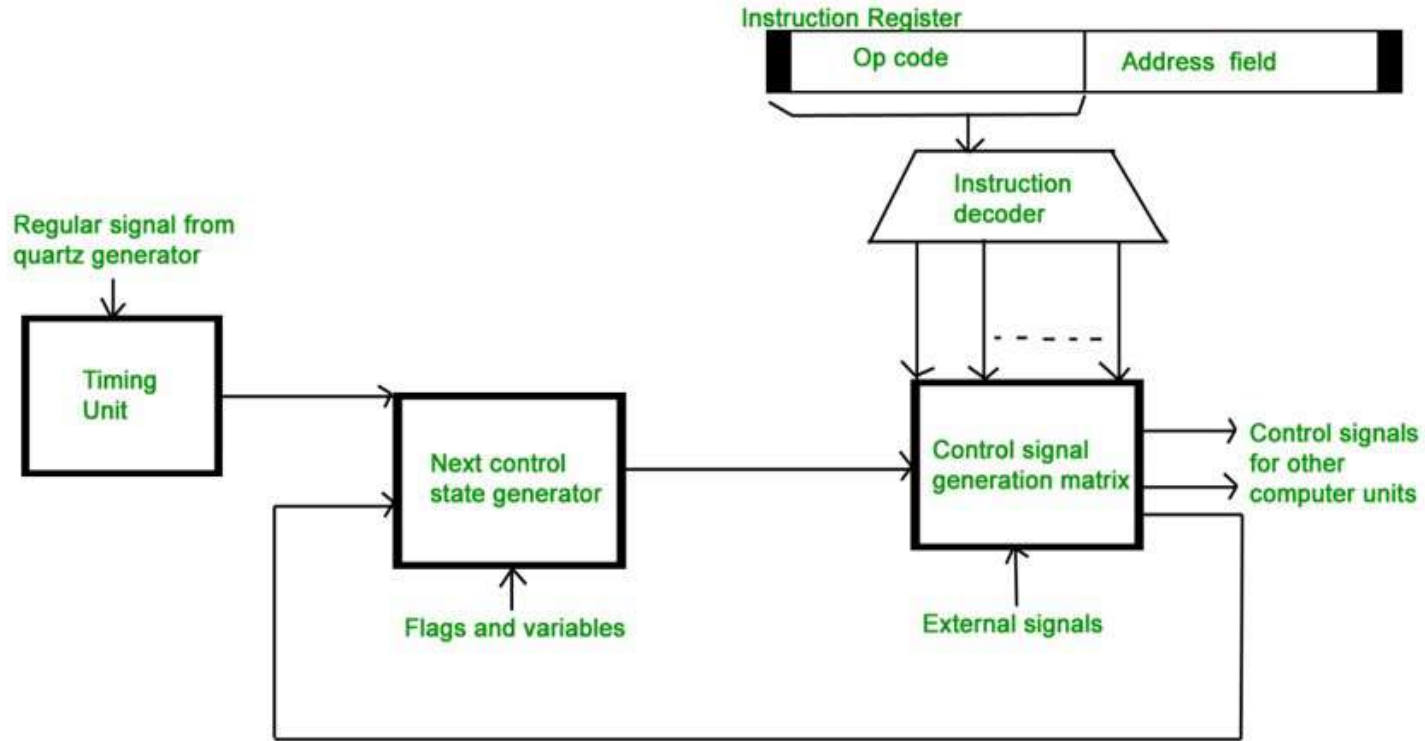
- A hardwired control unit is a method of generating control signals with the help of Finite State Machines (FSM).
- The control signals are generated by specially built hardware logical circuits.
- Signal production mechanism cannot be changed without physically changing the circuit structure.

Characteristics of Hardwired Control Unit ⁵

- Instruction Register (IR): Holds the current instruction.
- Control Logic Gates: Decodes opcode and generates control signals.
- Sequence Counter (SC): Keeps track of the step in the instruction cycle (T₀, T₁, T₂, ...).
- Timing Generator: Generates clock pulses to increment SC.
- Control Signal Outputs: Activated based on current SC value and decoded instruction.



Working of Hardwired CU



Working of Hardwired CU

T0 – Fetch the Instruction

- $PC \rightarrow MAR$ (Program Counter to Memory Address Register)
- Read signal activated
- $PC \leftarrow PC + 1$ (increment PC)
- **SC = 0**

T1 – Read from Memory

- $MDR \leftarrow \text{Memory}[MAR]$ (data read from memory into Memory Data Register)
- **SC = 1**

T2 – Load Instruction

- $IR \leftarrow MDR$ (instruction loaded into Instruction Register)
- **SC = 2**

Working of Hardwired CU

T3 onwards – Decode & Execute

- Opcode in IR is decoded using combinational logic
- Control logic activates specific control signals based on the opcode
- SC steps advance through micro-operations like:
 - $A \leftarrow A + B$ for ADD
 - $MAR \leftarrow \text{Address for memory access}$
 - $AC \leftarrow AC \& B$ for AND
- **SC is incremented at every clock pulse** unless a reset signal is given.

End of Instruction

- After the last step, **SC is reset to 0** to fetch the next instruction.
- This marks the beginning of the next instruction cycle.

Timing Table for an Example

Instruction: ADD operand

Operation: Add the operand from memory to accumulator ($AC \leftarrow AC + M[\text{operand}]$)

Time Step	Operation	Control Signals Activated
T0	$MAR \leftarrow PC, PC \leftarrow PC + 1$	Load MAR, Increment PC, Enable PC out, MAR in
T1	Read Memory $\rightarrow$ MDR	Memory Read, MDR in
T2	$IR \leftarrow MDR$	MDR out, IR in
T3	Decode IR, $MAR \leftarrow$ operand	IR decoded, Load operand address into MAR
T4	Read Memory $\rightarrow$ MDR	Memory Read, MDR in
T5	$AC \leftarrow AC + MDR$	MDR out, ALU ADD, AC in

Control Signal Table for an Example

Instruction: ADD operand

Operation: Add the operand from memory to accumulator ($AC \leftarrow AC + M[\text{operand}]$)

Control Signal	Purpose	Active During
PC out	Sends PC contents to MAR	T0
MAR in	Loads value into MAR	T0, T3
PC increment	Increments PC to point to next instruction	T0
Memory Read	Triggers read from memory	T1, T4
MDR in	Loads memory data into MDR	T1, T4
MDR out	Outputs MDR data	T2, T5
IR in	Loads instruction into IR	T2
ALU Add	Performs addition	T5
AC in	Loads result into Accumulator	T5

Applications of the Topic

- High-Speed Processors used where speed is critical due to fast control signal generation.
- Embedded Systems where instructions are limited and don't change often like washing machines.
- RISC Architecture Implementation where the instruction set is simple, and hardwired control provides minimal latency in execution.
- Pipeline-Compatible CPUs employed using instruction pipelining, where deterministic, fast control signal generation is necessary to maintain pipeline flow and efficiency.

Summary of the Lecture

1. A Hardwired Control Unit generates control signals using fixed logic circuits such as gates, flip-flops, and counters.
2. It follows a predefined sequence (often using a sequence counter) to execute each instruction step-by-step with high speed.
3. It is commonly used in systems where fast execution and predictable control are crucial, such as in RISC processors, embedded systems.
4. While it offers high performance and simplicity, it lacks flexibility—modifying control logic requires hardware changes, making it less ideal for complex or evolving instruction sets.

Next Lecture

- **Micro-Programmed Control Unit**
- **Comparative study**

Quiz

Q1. What is the primary advantage of using a hardwired control unit in a CPU?

- A. Easy to modify instruction sets
- B. Simplifies the control logic
- C. Faster execution of control signals**
- D. Uses less combinational logic

Q2. In a hardwired control unit, control signals are generated by:

- A. Microinstructions stored in ROM
- B. Finite state machine and logic gates**
- C. A software-based controller
- D. Stack-based hardware architecture

Q3. Which of the following is a limitation of the hardwired control unit?

- A. High execution time
- B. Limited instruction set
- C. Difficult to update or modify**
- D. Requires microprogram memory

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COMPUTER ORGANIZATION & ARCHITECTURE

(25CST-201/25ITT-201)

Unit No. 2

Chapter No. 2.1

Lecture No. 2.1.2

Topic : CPU Stack Organization, RISC and CISC Architecture

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

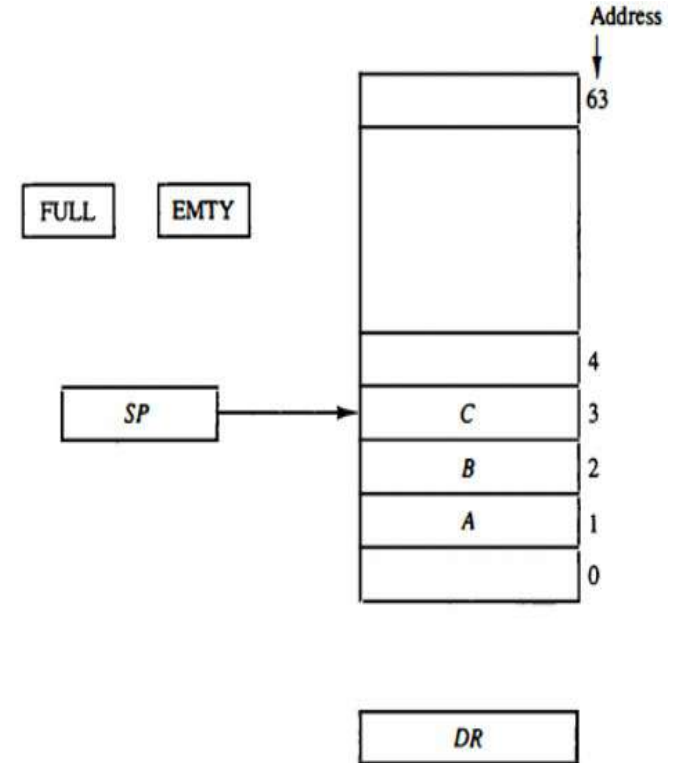
1. To understand the structure and working of stack-based CPU organization, where operands are implicitly stored and retrieved from a stack.
2. To learn the fundamental differences between RISC (Reduced Instruction Set Computer) and CISC (Complex Instruction Set Computer) architectures.
3. To analyze how stack organization and instruction set complexity impact CPU performance, instruction execution, and system design trade-offs.

Content

- CPU Stack Organization
- RISC Architecture
- CISC Architecture

Stack Organization

- The memory of a CPU can be organized as a STACK, a structure where information is stored in a Last-In-First-Out (LIFO) manner.
- Stack operations involve two main actions: Push and Pop.

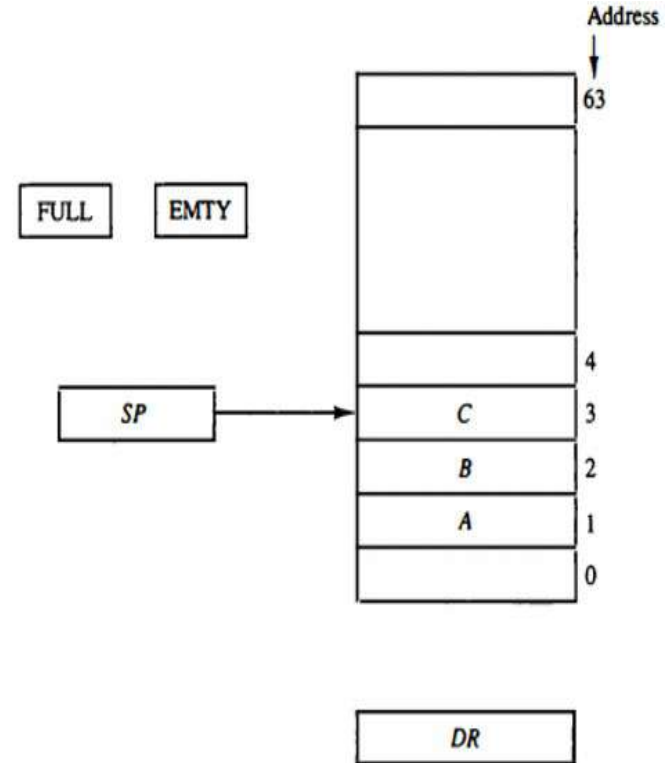


Stack Organization

- There are two main types of stacks:
 - Register Stack: Utilizes processor registers to create a stack structure, enhancing speed and efficiency in certain operations.
 - Memory Stack: Involves using dedicated memory locations to implement the stack structure.

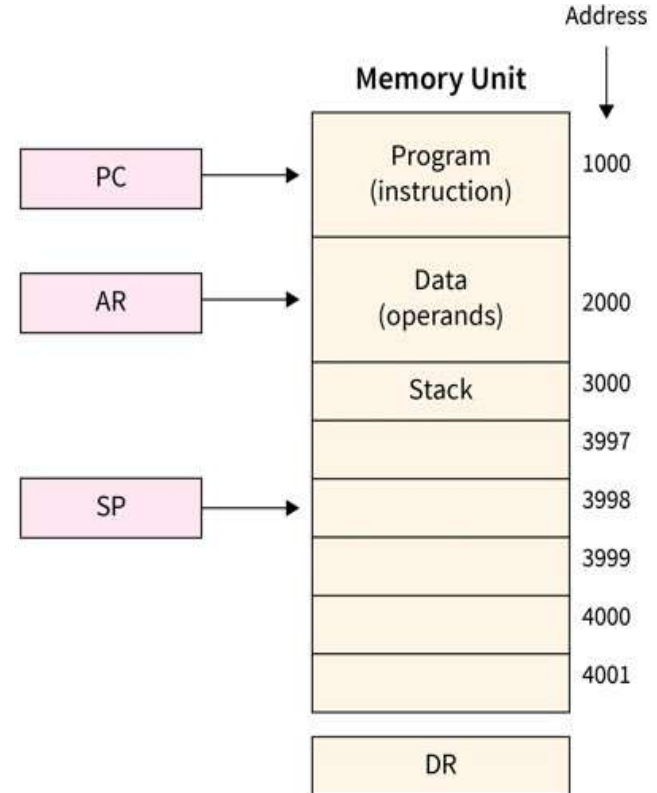
Register Stack

- When processor registers are organized in a stack-like fashion, it is termed a register stack.
- The stack pointer (SP) contains the address of the topmost element in the stack.
- When the stack is empty, EMPTY flag = 1, and when the stack is full, FULL flag = 1.
- The DR (Data Register) contains the data either being popped from or pushed into the stack.



Memory Stack

- When primary memory is organized in the form of a stack, it is referred to as a Memory Stack.
- PC indicates the address of the next instruction in the program.
- The Address Register (AR) points to an array of data within the memory stack.
- SP identifies the top of the stack.



RISC

- A reduced instruction set computer (RISC) is a type of microprocessor architecture that embraces simplicity and speed.
- The core philosophy is to optimize the execution of instructions by using a smaller, more straightforward set of commands.
- This enables faster instruction execution and facilitates pipelining, a technique where multiple instructions are overlapped in the execution process.

CISC

- Complex Instruction Set Computer (CISC) is a microprocessor architecture that attempts to improve performance by using complex instructions.
- It carries more complex operations within one instruction, reducing the number of instructions required per task, but using more than one clock cycle to execute an instruction.

Applications of the Topic

- Embedded Systems and Microcontrollers due to their simplified instructions, low power consumption, and faster execution.
- High-Level Language Execution via Stack CPUs ideal for executing expressions and function calls in languages like Java (via JVM) and Forth, where operand handling through stacks simplifies control flow.
- Desktop and Server Systems due to their rich instruction sets and backward compatibility with legacy software.
- Compiler and Virtual Machine Design that map high-level code efficiently onto hardware.

Summary of the Lecture

1. CPU stack organization uses a Last-In-First-Out (LIFO) approach for storing operands and results, simplifying instruction formats and minimizing the need for explicit operand addresses.
2. RISC focus on a small, efficient set of instructions that execute quickly and uniformly, ideal for high-speed and low-power systems.
3. CISC offer a rich set of instructions that can perform complex tasks in fewer lines of code, often used in general-purpose computing.

Next Lecture

- **Design of Control Unit**
- **Hardwired CU**
- **Micro-programmed CU**

Quiz

Q1. Which of the following best describes a characteristic of RISC architecture?

- A. Large number of complex instructions
- B. Variable-length instructions
- C. Few simple instructions with fixed length**
- D. Uses stack-based operand handling

Q2. In a stack-based CPU organization, operands for arithmetic operations are typically:

- A. Retrieved implicitly from the top of the stack**
- B. Specified explicitly in instruction
- C. Stored in general-purpose registers
- D. Fetched from memory addresses

Q3. Which of the following is a key advantage of CISC architecture over RISC?

- A. Lower instruction decoding complexity
- B. Faster execution speed for all tasks
- C. Uniform instruction size for better pipelining
- D. Fewer instructions per program due to complex instructions**

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Unit No. 2

Chapter No. 2.1

Lecture No. 2.1.1

Topic : CPU Organization-Accumulator Based, General Register Organization

Er. Shikha Atwal (E11186)

Assistant Professor

Learning Objectives

1. To understand the structural differences between accumulator-based and general register-based CPU organizations.
2. To learn how data operations are carried out using an accumulator versus multiple general-purpose registers.
3. To analyze the trade-offs in performance, flexibility, and instruction complexity between the two CPU organization types.

Content

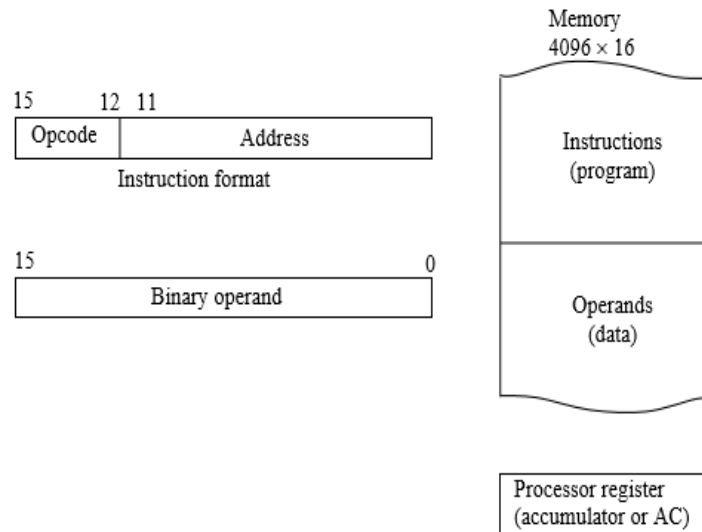
1. CPU Organization
 - 1.1. Accumulator Based
 - 1.2. General Register

CPU Organization

- Computer organization is concerned with the way the hardware components operate and the way they are connected together.
- CPU performs the following tasks:
 - Fetch instructions
 - Interpret instructions
 - Fetch data
 - Process data
 - Write data

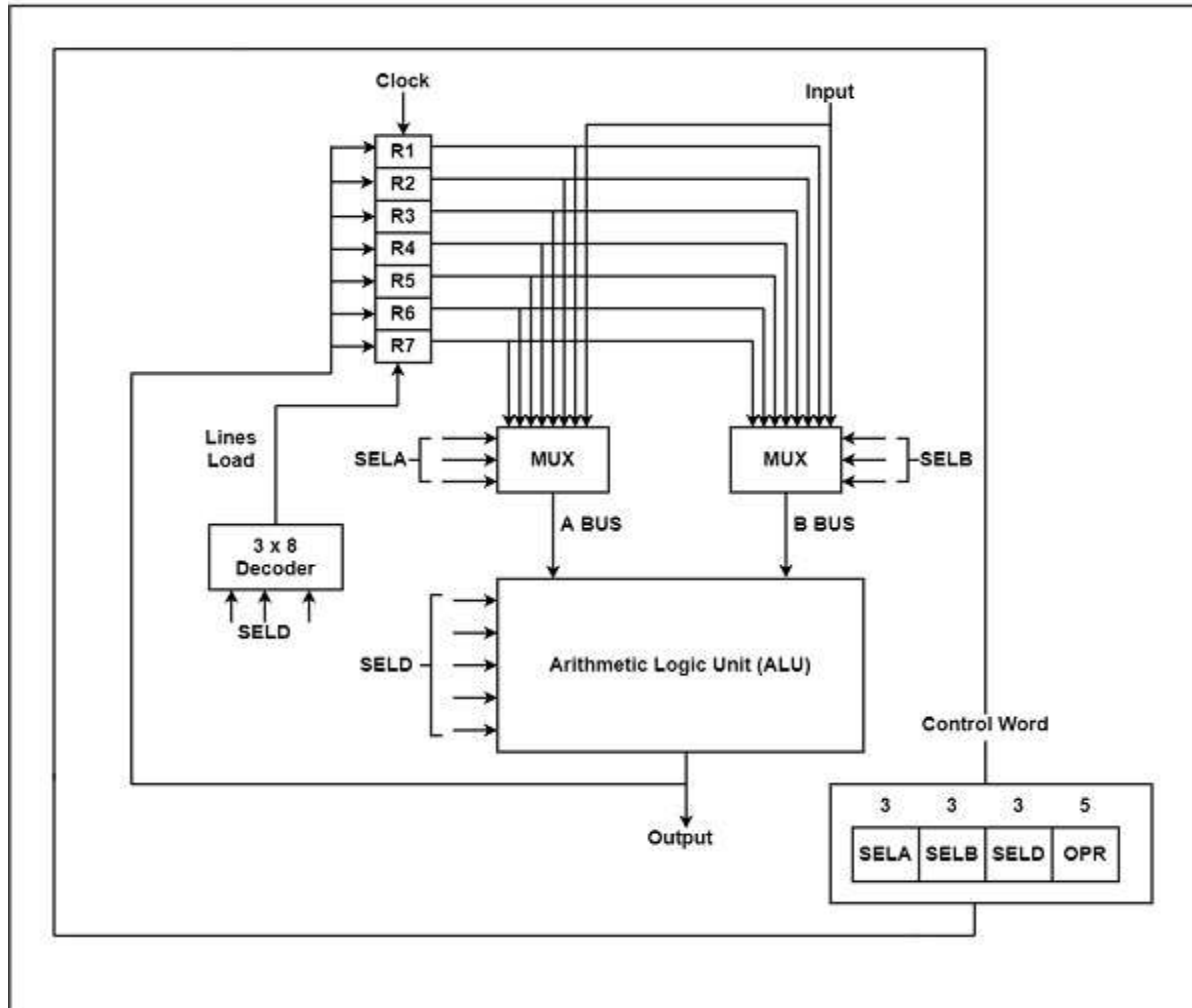
Single Accumulator Organization

- All operations are performed on an implied accumulator.
- Instruction format uses only one address field.
- Program Counter (PC) responsible for storing addresses, and it is connected to the memory.
- Additionally, there is an Accumulator (AC) connected to the ALU.

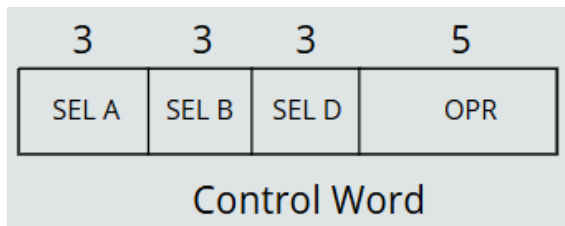


General Register Organization

- Instruction format needs 2 or 3 register address fields according to the operation.
- Generally CPU has seven general registers. Register organization show how registers are selected and how data flow between register and ALU.
- A decoder is used to select a 3 particular register. The output of each register is connected to two multiplexers to form the two buses A and B.
- The selection lines select the input data for the particular bus. The A and B buses form the two inputs of an ALU.
- The operation select lines decide the micro-operation to be performed.
- The result of the micro-operation is available at the output bus.



A control word, designed for the aforementioned CPU organization, consists of four fields as illustrated below:



Binary code	SELA	SELB	SELD
000	Input	Input	None
001	R1	R1	R1
010	R2	R2	R2
011	R3	R3	R3
100	R4	R4	R4
101	R5	R5	R5
110	R6	R6	R6
111	R7	R7	R7

The five bits of OPR define the type of operation to be performed by the ALU.

Operation Selection Code	Operation	Symbol
00000	Transfer A	TSF
00001	Increment A	INC
00010	A+B	ADD
00101	A-B	SUB
00110	Decrement A	DEC
01000	A AND B	AND
01010	A OR B	OR
01100	A XOR B	XOR
01110	Complement A	COM
10000	Shift Right A	SHR
11000	Shift Left A	SHL

Applications of the Topic

- Embedded Systems Design due to their minimal instruction set and reduced hardware complexity.
- Modern Processor Architectures for improved parallelism, speed, and efficient register allocation.
- Compiler Design and Optimization helps compilers generate optimized code based on available registers and data paths.
- Digital System Simulation and Teaching to demonstrate instruction execution and CPU internals.

Summary of the Lecture

1. CPU organization defines how data is processed and stored during instruction execution.
2. In an accumulator-based system, all operations involve a single register (the accumulator), making it simple and cost-effective.
3. A general register organization allows multiple general-purpose registers, enabling flexible operand handling, reduced memory access, and better performance.
4. This architecture is common in modern processors due to its support for faster computation and efficient compiler optimization.

Next Lecture

- **CPU Stack Organization**
- **RISC Architecture**
- **CISC Architecture**

Quiz

Q1. In an accumulator-based CPU, all operations are performed using:

- A. Memory directly
- B. Stack
- C. A single special-purpose register**
- D. General-purpose registers

Q2. Which of the following is an advantage of general register-based CPU organization over accumulator-based?

- A. Lower hardware cost
- B. Simpler instruction format
- C. No need for instruction decoding
- D. Reduced memory traffic and more flexible operations**

Q3. Which of the following processors is most likely to use accumulator-based organization?

- A. Intel Core i7
- B. ARM Cortex-M
- C. 8051 Microcontroller**
- D. AMD Ryzen

References/ Articles/ Videos

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- Vincent P. Heuring, and Harry F. Jordan, 'Computer Systems Design and Architecture', Second Edition, Pearson Education, 2003.

References/ Articles/ Videos

Other References

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Video References

- <https://youtu.be/k5YMLXPy1SE?si=NScdjrALFvXXkXc3>
- [Computer Organization #43: Accumulator Based Organization | CPU Organization | Practice Problem - YouTube](#)
- <https://youtu.be/Za7ozdjE8VI?si=4Ql6xAnNjODaPdgv>
- <https://youtu.be/vjqnWn5PdD0?si=CPEbp5esv0FpM-Ke>

Faculty-curated videos, NPTEL,¹⁵ Coursera, LinkedIn, or other relevant learning resources

- <https://www.coursera.org/learn/comparch>
- <https://www.coursera.org/learn/build-a-computer>
- https://www.youtube.com/watch?v=wjQQaY_gNG8
- <https://www.youtube.com/watch?v=SveHMRxl-al>
- <https://www.youtube.com/watch?v=1EA5qnHY9nY>



Thank You